# ASTRA environment-split clustering: cosmological forecasts with emulators and comparison with density split

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## Abstract

The distribution of galaxies holds more cosmological information than the standard two-point clustering statistics can capture: the non-linear cosmic web of nodes, filaments, walls and voids is also sensitive to the growth of structure, the matter content, dark energy and neutrinos. We ask how much of this extra information can be recovered with ASTRA, an algorithm that assigns an environment to every point from its neighbours in a Delaunay tessellation of the galaxies and of a random catalogue, without any smoothing scale. We split galaxies and random points by ASTRA environment and measure their auto- and cross-correlations, and compare them with the galaxy two-point function and with density-split clustering, the reference environment statistic. We build the covariances and cosmological derivatives needed for Fisher forecasts from three simulation suites: DESI-like galaxy mocks spanning 85 cosmologies (AbacusSummit), 1786 independent mocks of the official DESI programme, and the Quijote simulations. Along the way, we train a neural-network emulator of all the statistics and use it for an end-to-end inference test. ASTRA environment statistics constrain $\omega_b$, $\omega_{\rm cdm}$, $n_s$, $\sigma_8$, $w_0$ and $w_a$ about twice as well as the two-point function. They carry as much information as density split, and the two are complementary: combining them improves the constraints further. The original ASTRA cosmic-web classes, tested on Quijote, constrain $\sigma_8$ two to three times better than density split. We also describe the main difficulties of this kind of forecast (estimating derivatives from emulator suites, choosing a covariance, handling noisy derivatives) and how we dealt with them.

## 1 Introduction

Galaxy surveys such as DESI map the large-scale structure of the Universe over volumes large enough that statistical errors are no longer the main limitation. How the galaxies are distributed tells us how structures grew, how much matter there is, and what dark energy and neutrinos do. The question is how much of this information our statistics actually extract.

Most analyses compress the galaxy field into its two-point statistics, the correlation function or the power spectrum. For a Gaussian field this loses nothing, and on large scales it works very well. But gravity makes the late-time Universe strongly non-Gaussian: matter collapses into halos, gathers in filaments and nodes, and empties voids, and peculiar velocities distort the picture in redshift space. The two-point function does not see all of this. Statistics built on the cosmic web recover part of the missing information: [5] showed with Quijote that splitting the density field into nodes, filaments, walls and voids and combining the power spectra of these environments tightens the constraints substantially, most of all on the neutrino mass.

For galaxies, two families of environment statistics have been developed for this purpose. Density-split clustering [1, 2] ranks random points by the smoothed galaxy density around them, splits them into quantiles, and measures the auto-correlation of each quantile and its cross-correlation with the galaxies; combined with a simulation-based emulator [3], it improves ΛCDM constraints by factors of several over the two-point function. Void statistics, combined with galaxy clustering in a forward model of the galaxy–halo connection, also break degeneracies between cosmology and galaxy formation [6].

ASTRA [4] takes a different route. It triangulates the galaxies together with a uniform random catalogue and gives every point an environment from the fraction of galaxies among its Delaunay neighbours: there is no smoothing scale, and the environment adapts to the local density. ASTRA can classify points into voids, sheets, filaments and knots, or rank them continuously.

**Goal.** Our aim is to quantify how much ASTRA environment statistics improve the constraints on the cosmological parameters ($\Omega_m$ through $\omega_{\rm cdm}$, $\sigma_8$, the dark-energy parameters $w_0$ and $w_a$, and ideally the neutrino mass) over the standard galaxy two-point function, and how they compare with density split when both are measured on the same mocks, with the same random points and the same estimators. To this end we build dedicated covariance and derivative

datasets from suites of mock catalogues, which give Fisher forecasts of the parameter uncertainties, and we train a neural-network emulator of the statistics, which we use for an end-to-end inference test. The neutrino mass turned out to be out of reach with the available simulations (Section 7.2); all the other parameters are covered.

**Data.** We use three simulation suites, each for what the others cannot provide (Section 2): the AbacusSummit emulator suite of DESI-like galaxy mocks over 85 cosmologies, on which the measurements and the emulator are built; the official DESI mocks, whose 1786 independent boxes give a reliable covariance; and the Quijote simulations, whose paired simulations give clean derivatives. Getting reliable derivatives and covariances was the main difficulty of the work, and we describe how our choices were made and how the results evolved.

**Outline.** Section 3 presents the two environment estimators, Section 4 the clustering measurements, Section 5 the covariance, the derivatives and the Fisher formalism, Section 6 the successive analysis choices, Section 7 the results (DESI-like mocks, Quijote, ASTRA classes, inference), Section 8 the robustness tests, and Section 9 the discussion. The appendices explain how to reproduce the paper (A), give the mathematical details (B), and retrace the methodological path (C).

# 2 Simulations and mock catalogues

Table 1 summarises the data used, their versions and their role; exact paths, file names and the lists of catalogues are given in Appendix A.

## 2.1 AbacusSummit emulator suite

We use the HOD galaxy catalogues built for the DESI emulator programme on top of the AbacusSummit $N$-body suite [7]. Each simulation evolves $6912^3$ particles in a periodic box of $2\,h^{-1}$Gpc; we use the $z = 0.5$ snapshot. Halos are populated with luminous red galaxies (LRG) with the AbacusHOD model [8], whose 12 parameters (central and satellite occupation, velocity bias, environment-based assembly bias) are sampled in the prior of [9]. Every catalogue contains $4 \times 10^6$ galaxies, $\bar{n} = 5 \times 10^{-4}\,h^3\mathrm{Mpc}^{-3}$. The suite covers 85 cosmologies.

The catalogues are stored as `c{cosmo}_ph000/seed0/hod{NNN}.fits`. Two properties of this suite drive our methodology:

- **One phase per cosmology.** All cosmologies share the initial conditions `ph000`, with a single HOD seed.
- **One HOD draw per file.** The 500 files `hod000`–`hod499` of a cosmology have *different* HOD parameters (a Latin hypercube for emulator training), and the parameters behind a given index differ from one cosmology to another.

Different HOD files therefore cannot be used as independent realisations of the same model: the scatter between HOD draws reaches $\sim 55\%$ of the amplitude of the density-split monopoles, against $\sim 1\%$ for the cosmological response we want to measure.

**Cosmologies and parameters.** Besides the fiducial Planck 2018 ΛCDM cosmology (`c000`), we use the AbacusSummit linear derivative grid `c100`–`c113`: seven pairs, each varying one parameter ($\omega_b$, $\omega_{\rm cdm}$, $n_s$, $\alpha_s$, $w_0$, $w_0$ and $w_a$ together, $\sigma_8$) around the fiducial, with $h$ and $A_s$ adjusted to keep the angular size of the sound horizon $\theta_*$ and, for `c100`–`c105`, $\sigma_8$ fixed. Derivatives with respect to $\omega_b$, $\omega_{\rm cdm}$ and $n_s$ are therefore taken at fixed $\sigma_8$. All seven pairs are used to train and validate the emulator. The Fisher forecasts include the four ΛCDM parameters of Table 2, which are those in common with the density-split forecasts of [1] and were the only ones of the first version of this work, and, since the emulator and the derivative grid allow it, the dark-energy parameters $w_0$ and $w_a$: six cosmological parameters in total. Going from four to six parameters degrades the marginalised errors on $\omega_{\rm cdm}$ by $\sim 40\%$ and on $n_s$ by $\sim 15\%$, leaves $\omega_b$ and $\sigma_8$ unchanged, and changes the ASTRA/density-split ratios by less than their uncertainty. The remaining emulator parameters ($\alpha_s$, $N_{\rm ur}$) are fixed to their fiducial values in the Fisher matrices; all eight are free in the MCMC inference (Section 7.4).

**Positions.** The line of sight is one axis of the box ($z$ by default, $x$ and $y$ for the subvolume covariance). Redshift-space positions are `X_PERP`, `Y_PERP`, `Z_RSD`; real-space positions use `Z_PERP`. These columns already include the Alcock–Paczynski rescaling of the fiducial cosmology ($q_\perp$, $q_\parallel$, stored in the headers), so the box is $2000/q_\perp$ across and $2000/q_\parallel$ along the line of sight.

## 2.2 Official DESI AbacusHOD LRG mocks

The emulator suite contains a single phase per cosmology, so its only source of covariance is the 64 subvolumes of one box, measured along three correlated lines of sight (192 samples). This limits the length of

| Data set | Version / content | Used | Role |
|---|---|---|---|
| AbacusSummit emulator suite (DESI LRG HOD mocks) | AbacusSummit base boxes $2\,h^{-1}$Gpc, $z = 0.5$, AbacusHOD, `yuan23_prior`; 85 cosmologies, phase `ph000`, seed 0, 500 HOD draws each | $\simeq$ 1040 catalogues: 8 HODs per cosmology (678), 72 fiducial HOD-calibration, 240 near-fiducial (densification), 54 extra; 27 in real space | measurements, emulator training/validation/test, finite differences, regression, single-box covariance (`c000 hod239`) |
| Official DESI AbacusHOD mocks | `AbacusHOD_mocks/v1`, LRG, $z = 0.5$, model `velbias` (best fit to early DESI LRG data); `CovBox`: 1786 small boxes $500\,h^{-1}$Mpc (`ph3000+`); `CubicBox`: 25 base boxes `c000 ph000--024` | all 1786 small boxes and 25 base boxes | baseline covariance, fiducial point, MCMC test data |
| Quijote | $1\,h^{-1}$Gpc, $512^3$ particles, Rockstar halos, snapshot 3 ($z = 0.5$, `out_3_pid.list`) | 4000 `fiducial`; 500 each of `Om`, `Ob2`, `h`, `ns`, `s8` ($\pm$), `Mnu_p/pp/ppp`, `fiducial_ZA` | clean finite-difference Fisher forecasts |

Table 1: Data sets used in this work.

| Parameter | fiducial | + / − | step $\theta_+ - \theta_-$ |
|---|---|---|---|
| $\omega_b$ | 0.02237 | c100 / c101 | 0.00089 |
| $\omega_{\rm cdm}$ | 0.1200 | c102 / c103 | 0.0079 |
| $n_s$ | 0.9649 | c104 / c105 | 0.0200 |
| $\sigma_8$ | 0.808 | c112 / c113 | 0.0320 |

Table 2: AbacusSummit cosmologies of the derivative grid for the four ΛCDM parameters; the pairs `c106`–`c111` vary $\alpha_s$, $w_0$ and $(w_0, w_a)$.

the data vectors (Hartlap factor) and, as we found, biases the comparison of the statistics (Section 6). We therefore also use the official DESI AbacusHOD LRG mocks at $z = 0.5$ [8] (`AbacusHOD_mocks/v1`), generated with one best-fit HOD (model `velbias`, fitted to early DESI LRG clustering):

- the 1786 AbacusSummit *small* boxes (`c000`, phases 3000+), of side $500\,h^{-1}$Mpc and independent initial conditions;
- the 25 *base* boxes of `c000` (phases 0–24), of side $2\,h^{-1}$Gpc.

Choices. (i) Each small box is treated exactly like one subvolume of the main analysis: periodic images for the tessellation, density field smoothed on the periodic box, and the *open* Landy–Szalay estimator with geometry randoms for the clustering, so that the covariance corresponds to the same measurement. A periodic small box has no modes larger than its side, whereas a subvolume of a larger box is affected by them (super-sample covariance); the base boxes (64 subvolumes × 25 phases) allow this effect to be measured. (ii) Positions are used as provided, the $z$ coordinate including RSD; only the $z$ line of sight is available. (iii) The best-fit HOD lies inside the prior of the emulator suite and its number density (62 900 galaxies per small box) matches that of our catalogues (62 500 per subvolume), which matters for ASTRA (Section 8). We therefore take it as the fiducial point of the baseline forecasts and of the inference, with this covariance. These mocks contain a single cosmology: they do not provide derivatives.

### 2.3 Quijote simulations

The derivatives cannot be estimated cleanly by finite differences in the emulator suite, because the HOD differs between cosmologies. The Quijote simulations [12], used by [1] for density-split Fisher forecasts, provide what is missing: for each parameter, 500 pairs of simulations with ± variations of $\Omega_m$ ($\pm0.01$), $\Omega_b$ ($\pm0.002$), $h$ ($\pm0.02$), $n_s$ ($\pm0.02$), $\sigma_8$ ($\pm0.015$), and simulations with $M_\nu = 0.1, 0.2, 0.4\,$eV, all sharing the initial seeds of the corresponding fiducial realisations, plus thousands of fiducial realisations for the covariance. Each is a periodic box of $1\,h^{-1}$Gpc with $512^3$ particles.

Choices. (i) Tracers: all Rockstar host halos at $z = 0.5$ (no subhalos, at least 100 particles). Their number density, $n \simeq 5 \times 10^{-5}\,h^3{\rm Mpc}^{-3}$, is ten times lower than that of the LRG mocks and cannot be raised with these catalogues. Since ASTRA is an adaptive-scale statistic (Section 8), its effective scale, the distance between neighbours $\propto \bar{n}^{-1/3}$, is larger in Quijote ($\simeq 27\,h^{-1}$Mpc) than for the LRGs ($\simeq 13\,h^{-1}$Mpc). Both methods are measured on the same halos, so the comparison of ASTRA and density split remains like-for-like, but at another density: absolute errors and ratios should not be compared directly between the Quijote and AbacusSummit anal-

yses. (ii) Redshift space along $z$: $z_s = z + v_z/[aH(a)]$, with the $\Omega_m$ of each simulation. (iii) The whole box is one volume, with periodic images for the tessellation and the natural estimator of a periodic box (analytic random pairs) for the clustering; density split uses the same smoothing ($R_s = 10\,h^{-1}$Mpc) as for the LRGs. (iv) Derivatives are paired central differences averaged over the realisations shared by the + and − sets; for $M_\nu$, the three-point forward difference $(-3f_0+4f_{0.1}-f_{0.2})/0.2$ with the Zel'dovich-initialised fiducial set (`fiducial_ZA`) for $M_\nu = 0$, as is standard for Quijote. Because the tracers are the same halos in all simulations, these derivatives have no HOD mismatch. The price is realism: halos are not DESI LRGs, so Quijote serves as a controlled comparison of the information content of the two methods, complementary to the DESI-like analysis.

# 3 Environment estimators

Both estimators are evaluated on the *full periodic box*, and on the same set of $N_r = N_g$ uniform random points (fixed seed, drawn in the unit cube and rescaled to the box, so that the random points are identical in all cosmologies up to the AP rescaling).

## 3.1 ASTRA split

The galaxies and random points are joined into a single catalogue and Delaunay-triangulated. For every point (galaxy or random), with $N_{\rm data}$ and $N_{\rm rand}$ the numbers of galaxies and randoms among its Delaunay neighbours,

$$r = \frac{N_{\rm data} - N_{\rm rand}}{N_{\rm data} + N_{\rm rand}} \in [-1, 1]. \qquad (1)$$

A point surrounded by galaxies ($r \to 1$) lies in a dense region, a point surrounded by randoms ($r \to -1$) in an underdense one. In the original ASTRA classification [4], thresholds on $r$ define voids, sheets, filaments and knots; here we use $r$ as a continuous environment variable. The triangulation is computed independently in each subvolume (Section 4) on a region padded by $30\,h^{-1}$Mpc on each side, with periodic wrapping across the faces of the box, and $r$ is kept only for the points inside the subvolume. This removes the edge effects of a non-periodic triangulation (missing neighbours, elongated boundary simplices). The mean inter-point separation of the joint catalogue is $\simeq 10\,h^{-1}$Mpc, so the padding contains several layers of neighbours (convergence in Section 8).

## 3.2 Density split

As a reference we use the standard density split [1]. The galaxies are assigned to a periodic mesh of $5\,h^{-1}$Mpc cells with a cloud-in-cell (CIC) scheme, the overdensity field is smoothed with a Gaussian kernel of radius $R_s = 10\,h^{-1}$Mpc by FFT, and the smoothed density contrast $\delta_R$ is read (CIC) at the position of every galaxy and random point.

## 3.3 Environment quartiles

For each method, each subvolume and each population, points are ranked by their environment value ($r$ or $\delta_R$) and split into four quartiles of exactly equal size, Q1 (most underdense) to Q4 (most overdense). Because $r$ is a ratio of small integers, many points share the same value; ties are broken at random. We consider two populations, ranked separately so that the thresholds of one do not depend on the other:

- **random quartiles**: the random points ranked by their environment, the query points of density split, sampling environments by volume;
- **galaxy quartiles**: the galaxies ranked by their own environment, sampling environments by mass.

## 3.4 Cosmic-web classes and why we use quartiles

ASTRA was designed to classify points into four cosmic-web classes with fixed thresholds on $r$ [4]: voids ($r \le -0.9$), sheets ($-0.9 < r \le 0$), filaments ($0 < r \le 0.9$) and knots ($r > 0.9$). We use exactly the definition of the reference implementation (Eq. 1 with $N_r = N_g$, $r$ from a single random catalogue, these thresholds). On the DESI-like fiducial catalogue the classes are extremely unbalanced (Figure 3): 0.25% of the random points are voids, 65% sheets, 34% filaments and 0.05% knots; for the galaxies 0.07%, 49%, 51% and 0.3%. In Quijote, 0.7%, 72%, 27% and 0% of the random points. In practice the classification reduces to *two dominant classes*, sheets and filaments, i.e. to a split of $r$ at zero; the voids and knots contain a few hundred points per subvolume, too few for precise clustering measurements (Figure 1). This imbalance motivated the use of equal-size quartiles of $r$ in the main analysis, which keep the ranking of the environments by ASTRA while giving every environment the same statistical weight, as in density split.

The ASTRA paper reports 7–11% of random points in voids [4], against 0.25–0.7% here. Its samples are 11–73 times denser than our LRGs (number densities $1.7 \times 10^{-3}$ to $1.1 \times 10^{-2}\,{\rm Mpc}^{-3}$, against $1.5 \times 10^{-4}\,{\rm Mpc}^{-3}$). The fractions are not set by the density alone (the Quijote halos, ten times sparser than the LRGs but more strongly clustered, have more voids), but by the density contrast of the tracer

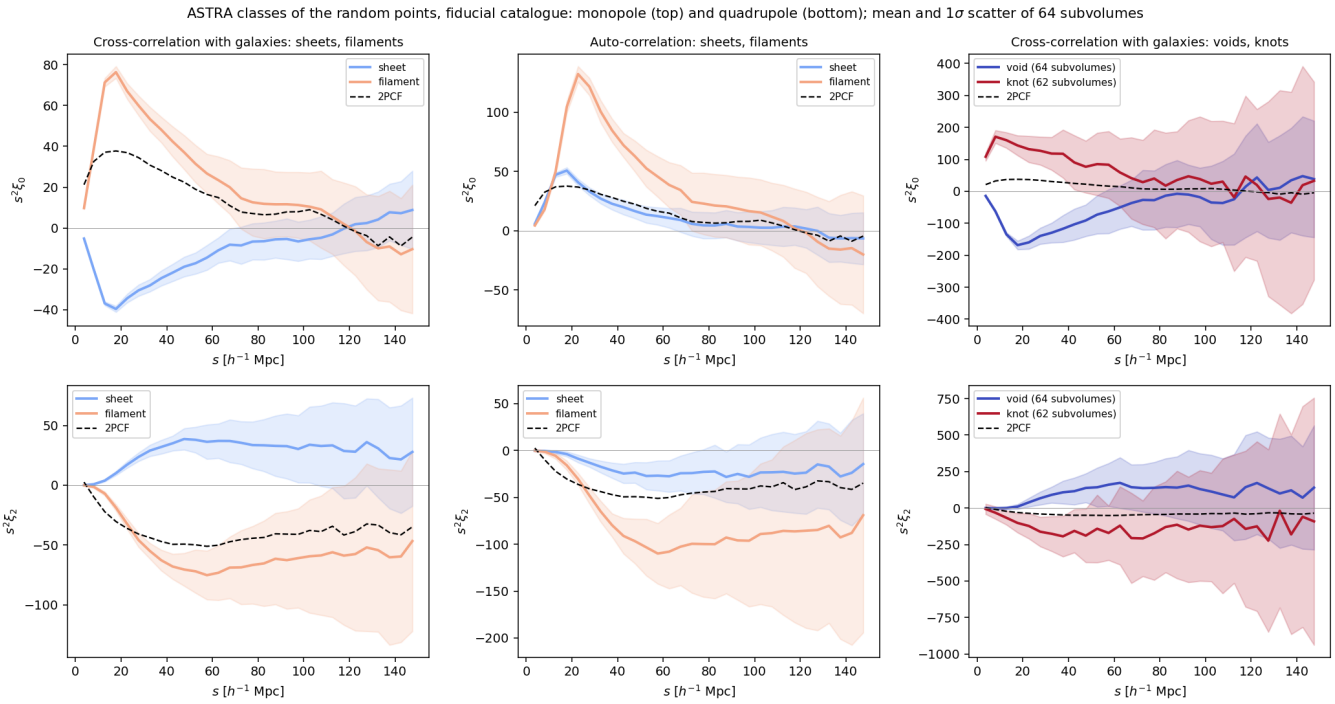



Figure 1: Monopole (top) and quadrupole (bottom) of the ASTRA classes of the random points on the fiducial catalogue: cross-correlation with the galaxies (left) and auto-correlation (centre) of the sheets and filaments, and cross-correlation of the voids and knots (right, different scale). Lines and bands: mean and $1\sigma$ scatter of the 64 subvolumes; dashed: 2PCF. The filaments behave like the overdense quartile Q4 and the sheets like the underdense quartile Q1; the voids and knots contain only $\sim 150$ and $\sim 30$ random points per subvolume, hence their large scatter.

field at the scale of the mean inter-point separation. This is another manifestation of the adaptive scale of ASTRA (Section 8): fixed thresholds on $r$ do not select the same environments in samples of different density and bias, whereas quantiles of $r$ do define comparable environments.

### 3.5 Comparison of the two classifications

Figure 2 compares the two estimators on the fiducial catalogue. We quantify their agreement with the Spearman rank correlation: the points are ranked by $r$ and by $\delta_R$, and the correlation coefficient of the two rankings is computed. It is 1 if the two methods order the points identically, 0 if their orders are unrelated, and it only depends on the ordering, which is what defines the quartiles. We find $0.719 \pm 0.005$ for the random points and $0.713 \pm 0.005$ for the galaxies (scatter over subvolumes): the two methods agree on the broad picture, dense or empty, but differ substantially point by point. The extreme quartiles are the most consistent: 63% of the ASTRA Q1 (Q4) random points are also in the density-split Q1 (Q4), whereas the intermediate quartiles are largely mixed, and less than 2% of the points move between the two extremes. ASTRA thus identifies the same large-scale structures as density split but redistributes many points (Figure 3), as expected from an adaptive rather than a fixed scale. The distributions of $r$ and $\delta_R$ per quartile are shown in Figure 4.

**Response to cosmology.** Figure 5 shows how the mean environment value of each quartile of the Quijote halos shifts when a cosmological parameter is varied, in units of its scatter between realisations. The density-split contrast $\delta_R$ responds strongly (up to $7\sigma$ for $\sigma_8$), with opposite signs for the $+$ and $-$ variations: a more clustered Universe makes dense regions denser and empty regions emptier, and $\delta_R$, a density contrast in absolute units, records it directly. The ASTRA value $r$ hardly moves ($< 1\sigma$). It is a ratio of neighbour counts normalised by the local random points: it is bounded between $-1$ and 1, it saturates in the most extreme regions, and because the neighbourhood adapts to the local density, a uniform increase of the contrast changes the ranking of the points much more than the value of $r$. Two consequences follow. First, since both methods use quartiles, i.e. ranks, neither of our statistics uses the environment values themselves; what they use is *where* the points of each environment are, and the clustering of these positions responds to cosmology for both methods (Section 7). Second, for ASTRA the distribution of $r$ itself is a poor cosmological probe, unlike the density distribution of density split: the infor-

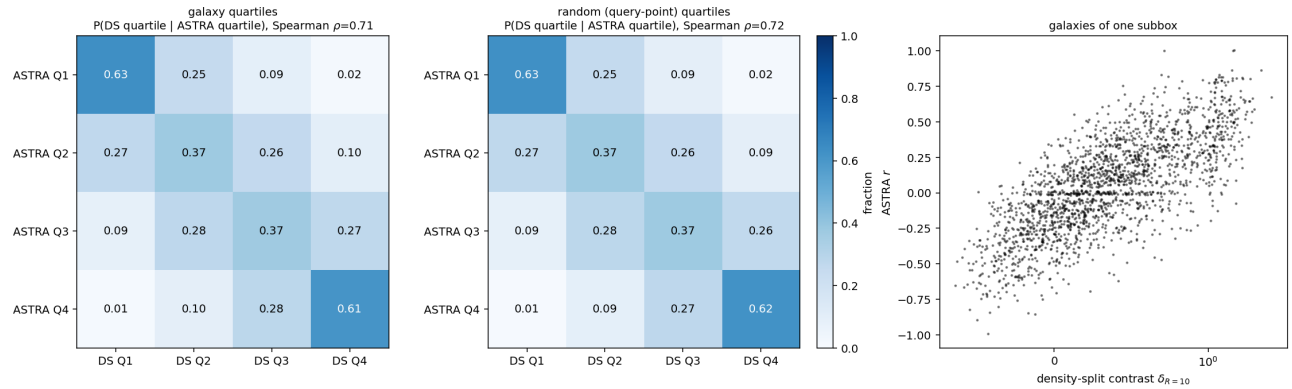



Figure 2: ASTRA and density-split environments on the fiducial catalogue (`c000`, `hod239`, line of sight $z$). Left, middle: probability that a point of a given ASTRA quartile falls in each density-split quartile, for galaxies and random points (average over 64 subvolumes). Right: ASTRA $r$ against the smoothed density contrast for a sample of galaxies (small vertical jitter added to the discrete $r$).

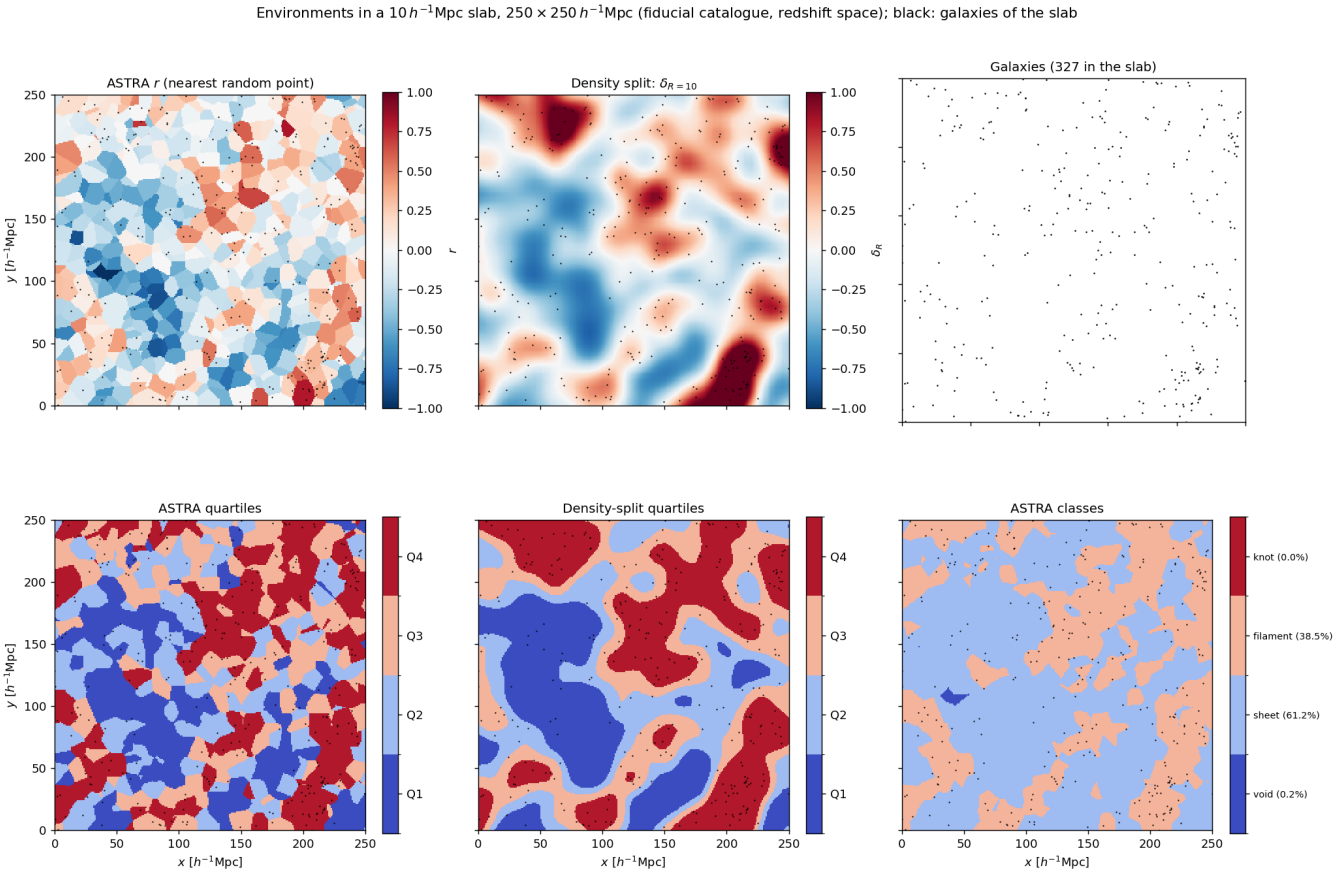



Figure 3: Environments in a $10\,h^{-1}$Mpc slab of $250 \times 250\,h^{-1}$Mpc of the fiducial catalogue, drawn as filled regions: each pixel takes the ASTRA value $r$, quartile or class of the nearest random point (left, bottom right), or the smoothed density contrast and its quartile (centre); black dots: galaxies of the slab. ASTRA defines environments cell by cell, at the scale of the Delaunay neighbours, whereas density split defines them on a smooth field; the ASTRA classes are dominated by sheets and filaments.

mation of ASTRA split lies in the clustering of its environments.

# 4 Clustering measurements

## 4.1 Subvolumes and estimator

Each $2\,h^{-1}$Gpc box is divided into $4^3 = 64$ non-overlapping cubic subvolumes of $500\,h^{-1}$Mpc (about 62 500 galaxies each). Each subvolume is treated as

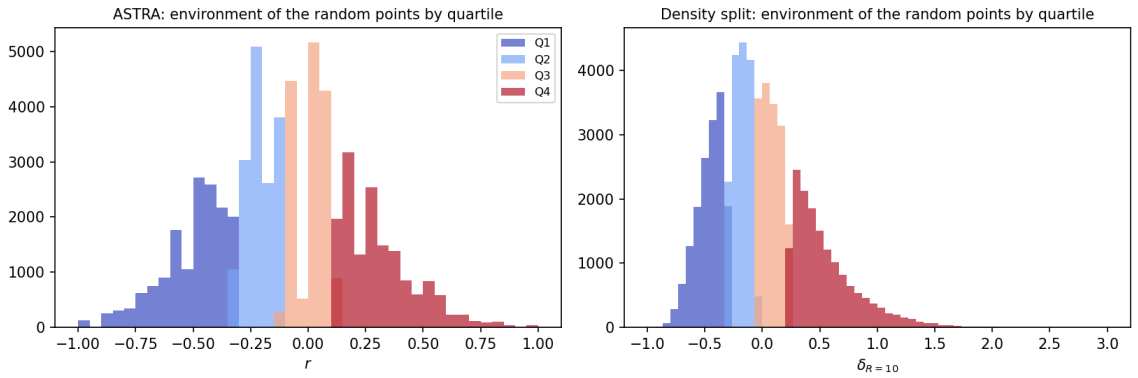



Figure 4: Distribution of the environment of the random points per quartile: ASTRA $r$ (left, discrete values) and density-split $\delta_R$ (right).

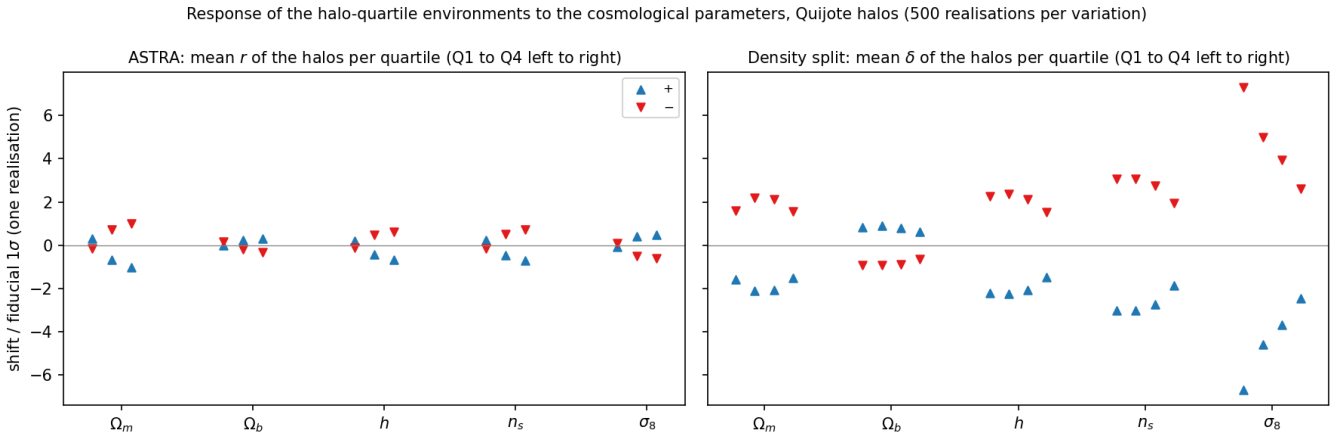



Figure 5: Shift of the mean environment of each halo quartile (Q1 to Q4 from left to right) for the + (blue) and − (red) variations of each parameter, Quijote, in units of the scatter of one realisation. Left: ASTRA $r$; right: density-split $\delta_R$.

an independent survey with open boundaries: correlation functions are measured with the Landy–Szalay estimator [10],

$$\xi(s,\mu) = \frac{D_1D_2 - D_1R - RD_2 + RR}{RR}, \qquad (2)$$

using a uniform catalogue of 5× the number of galaxies to describe the geometry (the same for all subvolumes, so that $RR$ is computed once). Here $D_1D_2$ counts the pairs of points of the two samples at separation $s$, $D_1R$ and $RD_2$ the pairs between each sample and the random catalogue, and $RR$ the pairs of random points: $RR$ gives the number of pairs expected in this geometry if there were no clustering, and the combination corrects for the edges of the subvolume. Pair counts are computed with `pycorr/Corrfunc` [11] in 30 bins of $5\,h^{-1}$Mpc in $s \in [0, 150]\,h^{-1}$Mpc and 120 bins in $\mu \in [-1, 1]$, with $\mu$ defined with respect to the fixed line of sight, and projected on the Legendre multipoles $\ell = 0, 2, 4$ (only $\ell = 0, 2$ are used in the forecasts).

## 4.2 Statistics

For each method $m \in \{\text{ASTRA, DS}\}$, population $p \in \{\text{random, galaxy}\}$ and quartile $\text{Q}_i$ we measure two statistics, together with the galaxy two-point correlation function $\xi_\ell^{\text{GG}}(s)$ (2PCF), the excess probability of finding two galaxies at separation $s$: 33 statistics per catalogue.

- The *quartile–galaxy cross-correlation* $\xi_{i,\ell}^{\text{QG}}(s)$, between the points of $\text{Q}_i$ and all the galaxies: the average density of galaxies at distance $s$ from the points of the quartile, i.e. the mean galaxy profile around these environments (negative at small $s$ around voids, positive around clusters).
- The *quartile auto-correlation* $\xi_{i,\ell}^{\text{QQ}}(s)$, between the points of $\text{Q}_i$ themselves: how the environments of one type are distributed in space relative to each other.

The first describes the shape of the structures, the second their large-scale distribution; both depend on cosmology. For random quartiles, they are the density-split cross- and auto-correlation functions of

[1]. For galaxy quartiles, which split the galaxies into four sets of exactly equal size, the average of the four quartile–galaxy cross-correlations is the 2PCF, up to a negligible difference in the normalisation of the pair counts: every galaxy pair counted in the 2PCF is counted once in one of the four cross-correlations, the four quartiles share the same random catalogue, and with equal sizes the normalised pair counts, and hence the Landy–Szalay estimator, which is linear in them, simply average. The quartiles therefore decompose the 2PCF by environment, and it is this decomposition that adds information.

### 4.3 Results on the fiducial cosmology

Figure 6 compares the extreme quartiles of the two methods, and Figures 7 and 8 show all four quartiles of each method in the layout of the density-split analyses [1]; the galaxy quartiles are shown in Figure 9.

The two methods produce qualitatively identical signals, which have a simple physical reading. The Q1–galaxy cross-correlation is negative at small separations: it is the galaxy profile around underdense regions, which contain fewer galaxies than average out to $\sim 50\,h^{-1}$Mpc. The Q4–galaxy cross-correlation is positive and above the galaxy 2PCF: around overdense regions the galaxy density is higher than around an average galaxy. The auto-correlations of the extreme quartiles are much larger than the 2PCF: extreme environments, both dense and empty, are strongly clustered among themselves, because they are not placed at random but trace the large-scale modes of the density field, which they sample with a large (positive or negative) bias. This amplification is why environment statistics are sensitive to the amplitude and shape of the large-scale fluctuations. The quadrupoles have opposite signs for Q1 and Q4: around underdense regions peculiar velocities point outwards and stretch the void profile along the line of sight (positive quadrupole), while around overdense regions infall compresses it (negative quadrupole); they measure the growth rate of structure. The intermediate quartiles Q2 and Q3 have weak cross-correlations of opposite signs, close to the mean field, and auto-correlations 5–6 times below those of the extreme quartiles: most of the signal comes from the extremes.

The ASTRA signals are systematically weaker, by a factor $\sim 1.6$ at the peak of the Q1 and Q4 cross-correlations ($s^2\xi_0 \simeq -90$ and $+90\,h^{-2}\mathrm{Mpc}^2$, against $-145$ and $+145$ for density split). The reason is that ASTRA estimates the environment of each point from only $\sim 15$ Delaunay neighbours, a noisy estimate: a point in a genuinely average region can, by chance, have mostly galaxy neighbours and be ranked in Q4. Each ASTRA quartile therefore contains a fraction of points that belong to other environments, which dilutes the contrast between quartiles (this is the mixing of Figure 2); density split averages over $\sim 10\,h^{-1}$Mpc, which reduces this noise. For density split the Q1 and Q4 auto-correlations are nearly equal, whereas for ASTRA the Q1 auto-correlation peaks 40% above the Q4 one: ASTRA's underdense quartile is relatively more clustered with itself, a first sign that its auto-correlations carry distinct information. A weaker signal does not mean less information: what matters is the response of the signal to the parameters relative to its noise, which the forecasts address.

### 4.4 Real space and redshift space

To isolate the effect of RSD, we also measured the fiducial catalogue (and the eight catalogues of the derivative pairs) in real space, the environments being then defined from the real-space positions (r-split) instead of the redshift-space ones (z-split), along the three lines of sight. Figure 10 compares the two. In redshift space, the monopoles at 20–$60\,h^{-1}$Mpc are larger, by a factor 1.6 for the galaxy 2PCF and by 1.3–1.5 for the Q1 and Q4 cross-correlations of both methods. This is the Kaiser effect: galaxies fall coherently towards overdense regions and flow out of underdense ones, which in redshift space compresses structures along the line of sight and makes them look more contrasted, so that the real-space amplitudes are lower. The same flows create quadrupoles that vanish in real space: positive around underdense regions (outflows) and negative around overdense ones (infall). In absolute value the quadrupoles of density split are 1.6–1.7 times larger than those of ASTRA split, following their larger monopoles. Relative to the monopole, however, the ASTRA quadrupole is the larger one ($|\xi_2|/|\xi_0| \simeq 0.8$ for ASTRA vs 0.6 for density split for Q1, 1.0 vs 0.8 for Q4, at 20–$60\,h^{-1}$Mpc). A plausible explanation is that ASTRA defines environments from the nearest neighbours in redshift space, at a scale ($\sim 13\,h^{-1}$Mpc) at which the distortions are strong, so that its selection of environments is itself anisotropic, whereas the isotropic smoothing of density split partly averages the distortions out. The ASTRA environments are thus less contrasted but more sensitive, relative to their amplitude, to the velocity field. Real-space environments are not observable, so all the forecasts use z-split; the real-space measurements are kept as a diagnostic. (A first version of this work used real space as a "truth" to quantify "systematic biases" of redshift-space forecasts; since both measurements and emulator are in redshift space, this comparison does not describe a bias of the analysis, and the relevant systematic, the

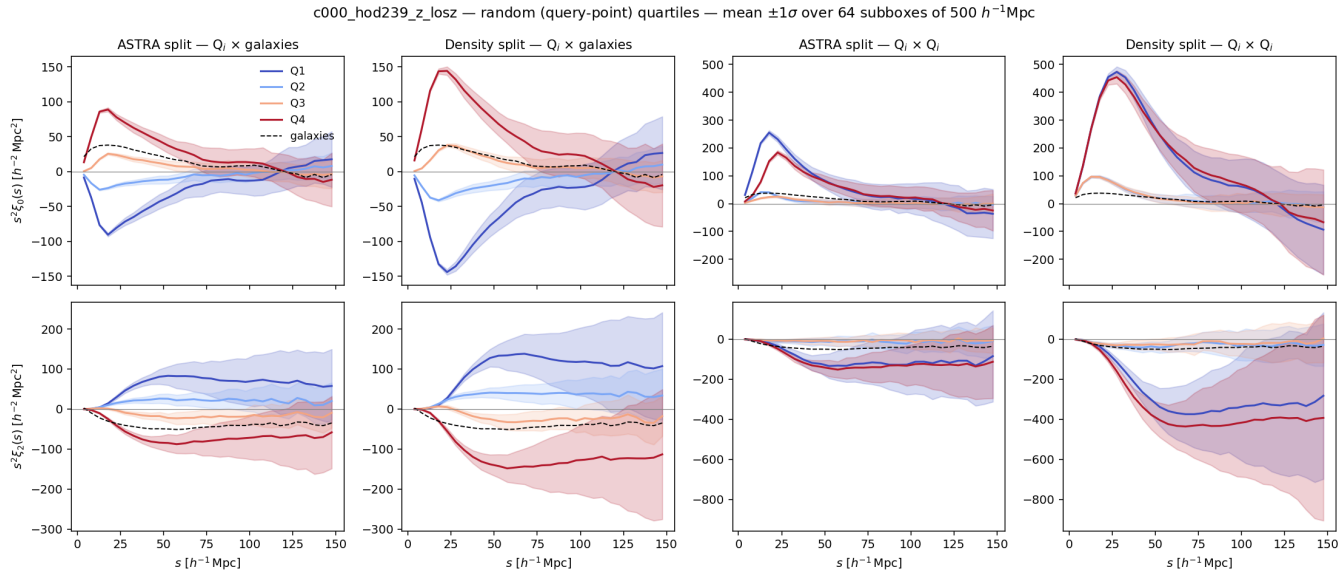



Figure 6: Monopole (top) and quadrupole (bottom) of the random-quartile statistics on the fiducial catalogue, for ASTRA split and density split: quartile–galaxy cross-correlations (left two columns) and quartile auto-correlations (right two columns), from Q1 (blue) to Q4 (red); dashed: galaxy 2PCF. Mean and $1\sigma$ scatter over the 64 subvolumes of $(500\,h^{-1}\mathrm{Mpc})^3$; panels of the same statistic share the same scale.

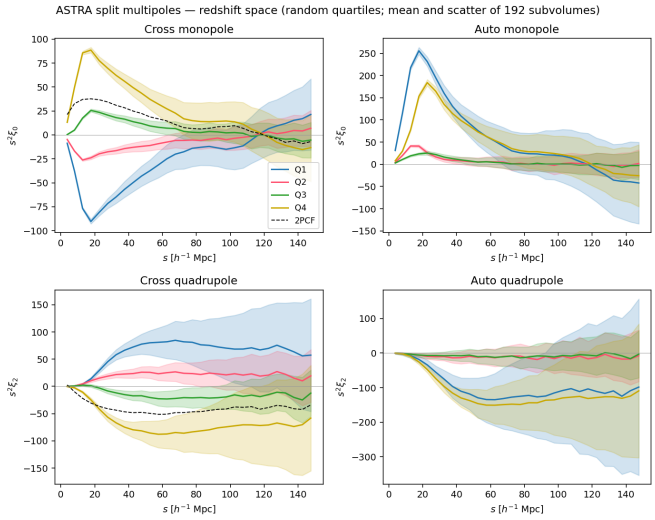



Figure 7: ASTRA split: monopole and quadrupole of the quartile–galaxy cross-correlations (left) and quartile auto-correlations (right) for the four random quartiles; dashed: galaxy 2PCF. Mean and scatter of 192 subvolumes of the fiducial catalogue.

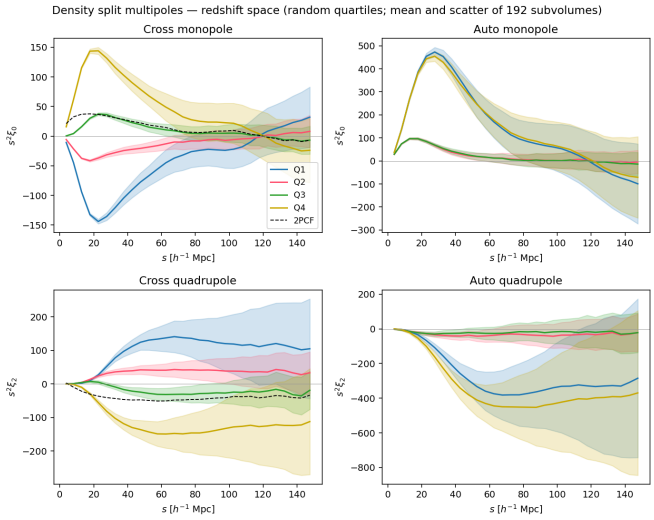



Figure 8: Same as Figure 7 for density split.

emulator error, is included in the covariance.)

## 5 Covariance, derivatives and Fisher forecasts

To forecast how well a statistic constrains the cosmological parameters, two ingredients are needed: how much the statistic fluctuates from one realisation of the Universe to another at fixed parameters, its *covariance*, and how much it changes when a parameter changes, its *derivatives*. A statistic is informative if its response to the parameters is large compared with its fluctuations. The Fisher matrix combines the two (Section 5.5). This section explains how we obtained each ingredient, and why each required a dedicated dataset.

**Data vectors.** By default, the cross-correlations of the extreme quartiles Q1 and Q4 with the galaxies, $\ell = 0, 2$, $20 \le s \le 150\,h^{-1}$Mpc in $10\,h^{-1}$Mpc bins (13 bins per multipole, 52 numbers per method), with or without the 2PCF (26 numbers). Variants add the quartile auto-correlations, use the galaxy quartiles, single quartiles, or change the minimum scale.

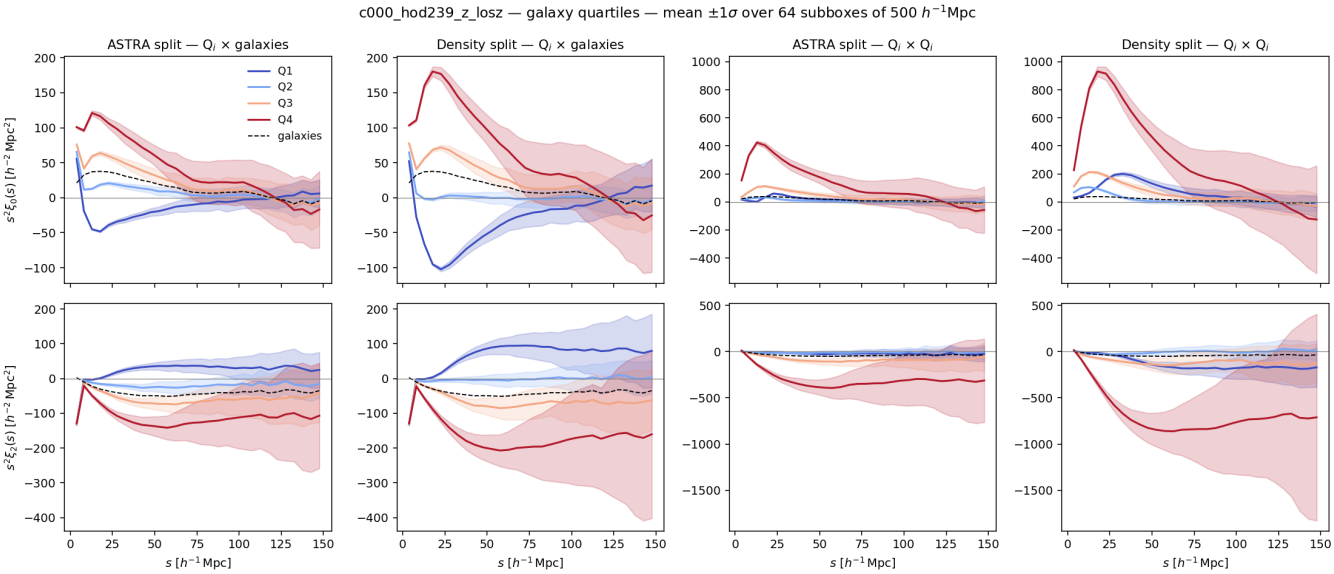



Figure 9: Same as Figure 6 for the galaxy quartiles.

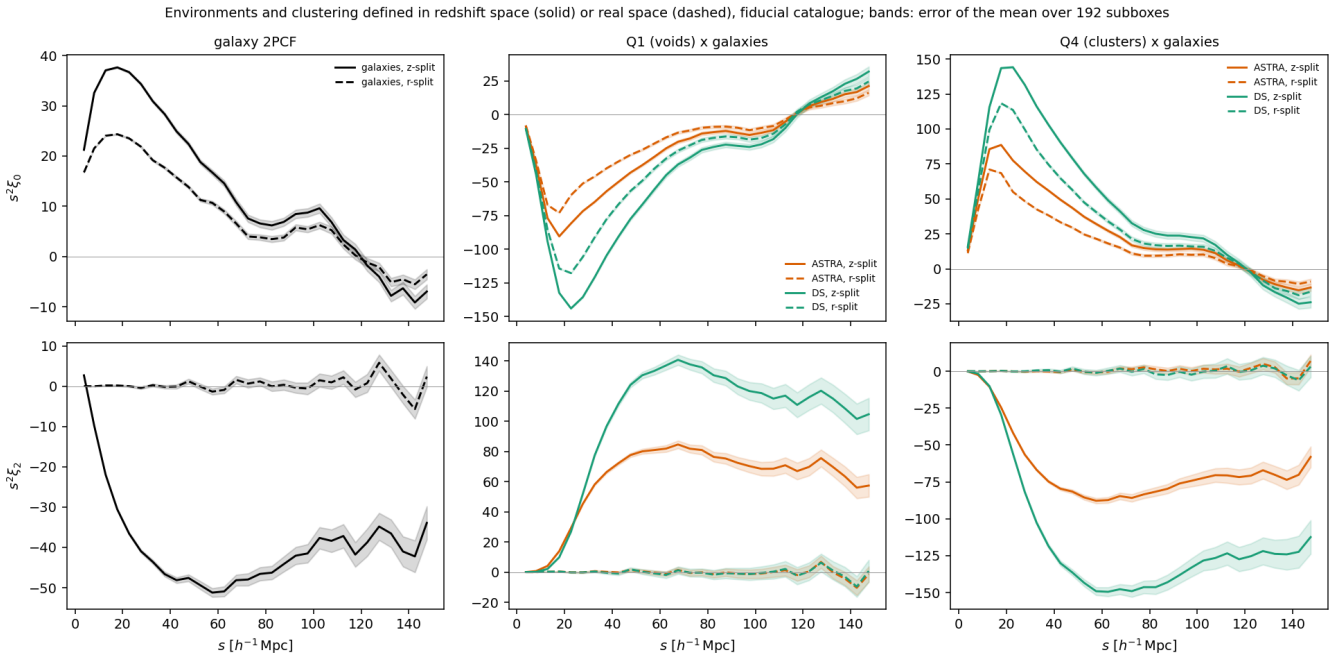



Figure 10: Environments and clustering defined in redshift space (solid) or real space (dashed) for the fiducial catalogue: galaxy 2PCF (left), Q1 (centre) and Q4 (right) random-quartile cross-correlations of ASTRA split (orange) and density split (green). Top: monopole, bottom: quadrupole. Bands: error of the mean over $3 \times 64$ subvolumes.

## 5.1 Covariance

The covariance sets the error bars of the forecasts and the correlations between the bins and statistics of a data vector. It must be estimated from many independent realisations of the same model, measured exactly like the data.

**Baseline: 1786 independent boxes.** The covariance is measured on the 1786 official small boxes (Section 2.2): 1786 independent $(500\,h^{-1}\mathrm{Mpc})^3$ volumes treated exactly like our subvolumes. Forecasts are quoted for this volume of $0.125\,(h^{-1}\mathrm{Gpc})^3$. The correlation matrix (Figure 11) shows strong correlations between neighbouring scales, between the Q4 cross-correlations and the 2PCF, and anti-correlations between the Q1 and Q4 blocks. The ASTRA and density-split blocks of the same quartile are very strongly correlated: being measured on the same galaxies, their noise is largely common. This is the origin of their complementarity: when two measurements share most of their noise but respond differently to the parameters, their combination cancels

part of that noise and gains much more than independent measurements would.

**First approach: 192 subvolumes of one box.** Before the official mocks were used, the covariance was estimated from the fiducial catalogue of the emulator suite (`c000`, `hod239`, the c000 draw closest to the centre of the HOD prior), using the 64 subvolumes measured along the three lines of sight, $N_s = 192$ samples. The three lines of sight are not independent, an approximation commonly used in simulation-based analyses. The inverse covariance is corrected by the Hartlap factor [14] (Appendix B), $(N_s - N_d - 2)/(N_s - 1)$: 0.72 for one method ($N_d = 52$), 0.59 with the 2PCF ($N_d = 78$), 0.31 for ASTRA + DS + 2PCF ($N_d = 130$), which motivated the compression of the data vectors (Section 5.5). Section 6 shows how replacing this covariance by the independent one changed the comparison.

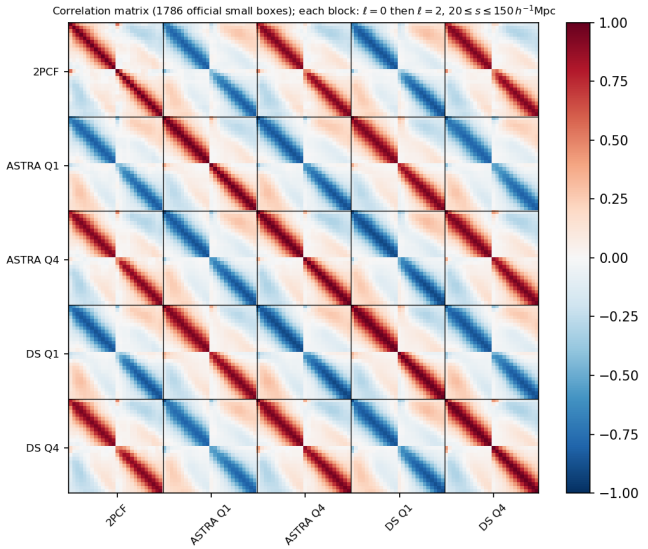



Figure 11: Correlation matrix of the 2PCF and of the Q1, Q4 cross-correlations of ASTRA split and density split (1786 official small boxes).

## 5.2 Finite differences: the HOD-mismatch problem

The AbacusSummit derivative grid provides, for each parameter, two simulations differing only by a small change of that parameter and sharing the same initial conditions. A natural estimate of the derivatives is the paired finite difference: for each parameter we select one HOD file in each cosmology of the pair, minimising the distance between the two HOD parameter vectors (Table 3), and difference the statistics subvolume by subvolume and line of sight by line of sight before averaging, so that the cosmic variance common to the two simulations largely cancels. This cancellation is only partial: each HOD catalogue is an independent draw of the galaxies populating the halos, so the galaxy sampling noise is not shared, and the difference of two full-box measurements with different HODs has an error comparable to that of a single box (1.05 times, median over the pairs of the grid). The precision of the paired derivatives comes from the average over the 192 paired samples.

| Parameter | HOD files | $\lvert\Delta h\rvert$ | random pairs |
|---|---|---|---|
| fiducial | c000_hod239 | – | – |
| $\omega_b$ | c100_hod448 / c101_hod694 | 0.63 | 1.38 |
| $\omega_{\rm cdm}$ | c102_hod589 / c103_hod841 | 0.58 | 1.38 |
| $n_s$ | c104_hod308 / c105_hod735 | 0.65 | 1.37 |
| $\sigma_8$ | c112_hod230 / c113_hod204 | 0.65 | 1.39 |

Table 3: HOD catalogues of the finite-difference pairs. $|\Delta h|$ is the distance between the two HOD parameter vectors normalised to the prior range (12 parameters); the last column gives the median for random pairs.

Averaged over the 192 samples, these differences have small statistical errors, but they are not cosmological derivatives, and this is one of the main findings of this work. A pair difference is the sum of two terms: the cosmological term, the change of the statistics due to the parameter (physics), and the HOD term, the change due to the different galaxy–halo connections of the two catalogues (a change in the galaxy-formation model of the mock, not in cosmology). We estimated the HOD term from 72 additional catalogues of the fiducial cosmology with different HODs, which give the response of the statistics to each HOD parameter, evaluated at the HOD difference of each pair, and compared it, in units of the statistical error, with the cosmological term predicted at fixed HOD: the HOD term is 10–15 times larger. The derivative vectors with respect to $\omega_b$, $n_s$ and $\sigma_8$ are then almost collinear (correlations 0.95–0.99 in the Fisher metric), because they all mostly measure the same HOD response, and the Fisher matrices are not positive definite. We tried to subtract the HOD term: a linear model of the HOD response predicts left-out catalogues with a 45% error, a Gaussian-process model (one GP per principal component of the data vector, anisotropic kernel) with 8%. But a correction that is 8% wrong on a term ten times larger than the signal leaves a residual of $0.08 \times 10 \simeq 0.8$ times the signal: the corrected derivatives are still dominated by the error of the correction, and the Fisher matrices remain unstable. Any emulator suite in which the HOD is drawn independently in each cosmology has this problem.

These finite differences remain useful in three ways. First, they measure the problem: they show by how much the HOD term dominates, which tells us that

any derivative from such a suite must be taken at fixed HOD. Second, once corrected for the HOD term with the Gaussian process, they give derivatives that do not involve a neural network, against which the emulator can be checked (Figure 14). Third, with several HOD draws in each cosmology of the derivative grid, they become a regression that estimates the derivatives at exactly the fiducial HOD (Section 5.4).

### 5.3 Neural-network emulator

**Why an emulator.** As in SUNBIRD [3], we model each statistic as a smooth function of the cosmological and HOD parameters, and take its derivatives at exactly fixed HOD by automatic differentiation, which removes the HOD mismatch by construction.

**Inputs and outputs.** The inputs are the 8 cosmological parameters varied in AbacusSummit ($\omega_b$, $\omega_{\rm cdm}$, $\sigma_8$, $n_s$, $\alpha_s$, $N_{\rm ur}$, $w_0$, $w_a$; $h$ and $A_s$ follow from $\theta_*$ and $\sigma_8$) and the 12 HOD parameters, rescaled to $[0,1]$ over the prior. The output of one network is one statistic, the monopole and quadrupole for $20 \le s \le 150\,h^{-1}$Mpc in $10\,h^{-1}$Mpc bins (26 numbers), measured on the full box as the mean of the 64 subvolumes, standardised by its mean and by the statistical error of the full box $\sigma_{\rm box}$. There is one network per statistic (33 statistics).

**Architecture and training.** Each network is a fully-connected multilayer perceptron: 20 inputs, three hidden layers of 256 units and 26 outputs, about $1.4 \times 10^5$ weights. This is the type of network used by SUNBIRD [3] for density-split statistics; the outputs are smooth functions of the inputs, which such networks represent well, and the size is a compromise between flexibility and the $\sim$ 1000 training catalogues. We did not explore other architectures systematically; the learning curve below indicates that the accuracy is limited by the noise of the training catalogues rather than by the network. We use SiLU activations rather than the common ReLU because they are smooth: the derivatives of a ReLU network are piecewise constant, which would make the Fisher derivatives noisy. The networks are trained with the Adam optimiser, with early stopping when the error on the validation cosmologies stops decreasing. With the standardisation above, the loss is a $\chi^2$ per bin, so that the network is penalised in proportion to the statistical significance of its errors. Each statistic has an ensemble of 10 networks with different initialisations; the prediction is the ensemble mean, and the scatter between members measures the uncertainty due to training.

**Training set.** 999 catalogues of the 85 cosmologies: 8 HOD draws per cosmology, the 72 fiducial calibration catalogues, and 16 draws close to the fiducial HOD in c000 and in each of the 14 cosmologies of the derivative grid, where the derivatives are evaluated (densification, below). Validation and test sets are 8 cosmologies each, never seen in training.

**Accuracy.** On the test cosmologies the emulator reproduces all 33 statistics with a median error of 0.4–0.9 $\sigma_{\rm box}$, the statistical error of a full $2\,h^{-1}$Gpc box (Figure 12). The errors are largest at the smallest scales and in the monopoles. What this means depends on the volume analysed: it is about ten times below the error of the $(500\,h^{-1}{\rm Mpc})^3$ volume of the forecasts, so for our comparisons the emulator error is negligible (it is nevertheless added to the covariance); for a full $(2\,h^{-1}{\rm Gpc})^3$ box it is comparable to the statistical error, so an analysis of such a volume, or of a DESI-like volume, would be limited by the emulator. A learning curve, retraining on 25, 50, 75 and 100% of the training set, shows that this error has reached a plateau at $\simeq 0.5\sigma_{\rm box}$, set by the galaxy sampling noise of single catalogues: more catalogues of the same kind would barely help, and lowering it requires several HOD realisations per training point (Appendix C).

**Cosmological response.** Accurate predictions do not guarantee accurate derivatives. We test the latter by training the emulator without the 14 cosmologies of the derivative grid and comparing its prediction of the $\pm$ pair differences, HOD mismatch included, to the simulations (Figure 13). The response to $\sigma_8$ and $n_s$ is reproduced to 10–20%, the response to $\omega_b$ to 30–45% with the correct shape, but the response to $\omega_{\rm cdm}$ is not captured below $s \simeq 70\,h^{-1}$Mpc: at fixed $\theta_*$ and $\sigma_8$ its effect is smaller than the emulator error and is not learnt away from the training points. This led us to *densify* the training set around the fiducial point with the 16 HOD draws closest to the fiducial HOD in c000 and in each cosmology of the grid (240 catalogues), so that the emulator sees the response to each parameter for many similar galaxy–halo connections. Densification alone did not stabilise the derivatives: the Fisher errors of the individual networks scattered by 70–170% for the 2PCF and for $\omega_{\rm cdm}$ and $n_s$, and the ASTRA/density-split comparison changed sign between trainings. As an independent check, the derivatives of the emulator at fixed HOD agree with the GP-corrected pair differences to 1–3% for $n_s$ and $\omega_{\rm cdm}$, 5–6% for $\sigma_8$ and 8–13% for $\omega_b$ (Figure 14).

**Training on catalogue differences.** To constrain the response near the fiducial point directly, we add to the loss the $\chi^2$ of the predicted *differences* between catalogues, over 2544 pairs: all the pairs $(c_+, h_i), (c_-, h_j)$ of the seven $\pm$ pairs of the grid with HODs within one prior range of the fiducial, and pairs of the same cosmology (HOD response); details in Appendix B. The differences are not much more precise than individual measurements, since the galaxy sampling noise is not shared, but the term constrains their *structure*: the network must reproduce how the statistics change between nearby points of parameter space, which is what the derivatives require. We also tested an additional weight $(25\,h^{-1}\mathrm{Mpc}/s)^\alpha$, $\alpha = 1$, on the small scales. We validated the three variants on 56 local catalogues (4 per grid cosmology) excluded from training, by predicting the differences of all the held-out $\pm$ pairs (Table 4, Figure 15). The difference term halves the error for $\omega_b$ and $\sigma_8$ and reduces it for all parameters but $\omega_{\rm cdm}$, already reproduced to 11% near the fiducial; the residuals reach the noise level of the measured differences ($\simeq 1\sigma_{\rm box}$), so the data do not allow a more stringent test. The small-scale weight brings a marginal additional gain but makes the forecasts that include the quartile auto-correlations unstable, so we use the difference term without it: this is the baseline emulator.

| | $\omega_b$ | $\omega_{\rm cdm}$ | $n_s$ | $\alpha_s$ | $w_0$ | $w_0, w_a$ | $\sigma_8$ |
|---|---|---|---|---|---|---|---|
| no diff. term | 21 | 11 | 35 | 14 | 17 | 17 | 13 |
| diff. term | 10 | 11 | 27 | 10 | 12 | 10 | 7 |
| + small scales | 8 | 11 | 26 | 9 | 11 | 8 | 7 |

Table 4: Median relative error (%) of the emulator prediction of the differences between held-out local catalogues of the $\pm$ pairs of the derivative grid, for the three training variants.

**Derivatives.** Figure 16 shows the baseline derivatives of the 2PCF and of the Q1, Q4 cross-correlations, as the change of each statistic for a small change of each parameter, in units of the statistical error. Three features matter for the forecasts. (i) The two methods respond very similarly to $\omega_{\rm cdm}$ and $n_s$, mostly through the monopole at $s < 75\,h^{-1}$Mpc: these parameters change the shape of the clustering as a function of scale. (ii) The Q1 (void) cross-correlations respond with the opposite sign to Q4 and to the 2PCF. A single statistic cannot tell a change of cosmology from, e.g., a change of galaxy bias, which also rescales the amplitude; two statistics that respond in opposite directions to the parameter but in the same direction to the bias can, which is how environments break degeneracies. (iii) At fixed HOD, $\sigma_8$ acts mostly on the quadrupoles, i.e. through the velocity field. Figures 17 and 18 show the response of the four quartiles: it is ordered from Q1 to Q4 for $\omega_{\rm cdm}$, $n_s$ and $w_a$, the extreme quartiles responding most, while the monopole responses to $\sigma_8$ and $w_0$ are small and have no simple ordering.

### 5.4 Derivatives from a regression over pairs

As an alternative to the emulator, we estimate the derivatives locally from the derivative grid alone. For a pair of cosmologies $c_\pm$, all the catalogues of each cosmology within a distance $r_{\rm max}$ of the fiducial HOD $h_0$ (the 16 draws of the densification, the pair draw and the emulator draws) are fitted with a first-order expansion in the HOD parameters, $\boldsymbol{d}_c(h) = \boldsymbol{a}_c + G\,(h - h_0)$, with a response $G$ shared by the two cosmologies; the derivative is $(\boldsymbol{a}_{c_+} - \boldsymbol{a}_{c_-})/(\theta_+ - \theta_-)$, the difference between the two cosmologies at exactly the fiducial HOD. The fit is performed subvolume by subvolume: the catalogues of the two cosmologies share the same initial conditions, so the same subvolume contains the same structures in both, and their difference is free of most of the cosmic variance. The errors are obtained by bootstrapping the subvolumes (refitting on random resamplings of them), which propagates the scatter between subvolumes into the derivatives. The intercepts are interpolated in the middle of draws surrounding $h_0$, and curvature common to the two cosmologies cancels in the difference. This estimator relies on a linear model and on the derivative grid only, while the emulator is a non-linear model of the whole suite: agreement between the two is a strong test of the derivatives.

**Reduced HOD space.** With 12 HOD parameters and $\sim 18$ catalogues per cosmology, the fit is poorly conditioned: there are almost as many unknowns as data points, so the fit can reproduce the catalogues it is given but not predict new ones. Its leave-one-out error (each catalogue predicted by a fit to the others) is 0.8–1.0 times the scatter between HODs, i.e. the fit predicts barely better than the mean. This is a classical overfitting problem, solved by reducing the number of free directions. We determine them independently, from the 72 fiducial calibration catalogues: a singular value decomposition of the linear HOD response of the statistics (each weighted by its statistical error) gives the combinations of HOD parameters ordered by how much they change the statistics. The singular values are $1, 0.054, 0.014, 0.013, \ldots$: *the clustering statistics respond essentially to a single combination of the 12 HOD parameters*, plausibly the large-scale galaxy bias, the others being 20 to 100 times weaker. Keeping 3 or 4 directions, the leave-one-out error drops to 0.7–0.9 times the HOD scatter,

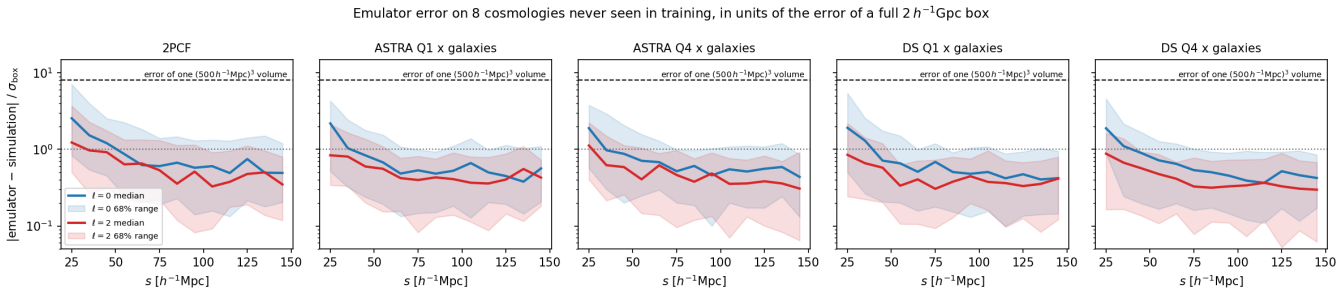



Figure 12: Emulator accuracy on 8 held-out cosmologies (8 HOD draws each): median (lines) and 68% range (bands) of |emulator − simulation| in units of the statistical error of a 2 $h^{-1}$Gpc box, for the monopole (blue) and quadrupole (red) of the 2PCF and of the Q1, Q4 cross-correlations of ASTRA split and density split. Dashed: error of one (500 $h^{-1}$Mpc)$^3$ volume, the volume of the forecasts.

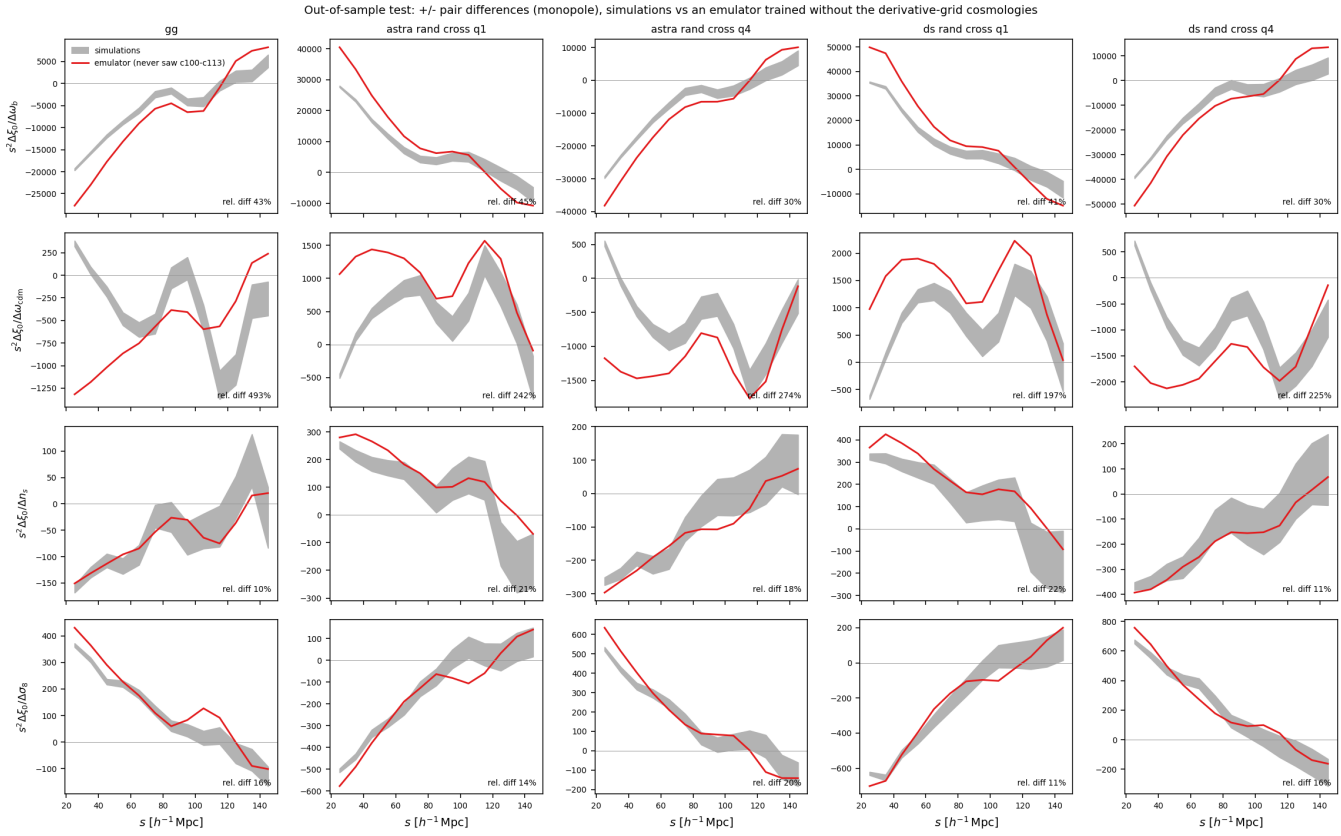



Figure 13: Out-of-sample test of the cosmological response: monopole of the ± pair differences in the simulations (grey band: mean ± error over the 192 paired samples) and predicted by an emulator trained without the cosmologies of the derivative grid (red).

at the level of the measurement noise, and the two choices give the same forecasts within 5%. We use 3 directions.

## 5.5 Fisher matrices

The derivatives $D = \partial \boldsymbol{d}/\partial(\boldsymbol{\theta}, \boldsymbol{h})$ are obtained at the fiducial point for the 6 cosmological parameters $\boldsymbol{\theta} = (\omega_b, \omega_{\rm cdm}, n_s, \sigma_8, w_0, w_a)$ and the 12 HOD parameters $\boldsymbol{h}$. The covariance is the sum of the measured covariance and of the (diagonal) emulator error: the emulator error is an additional uncertainty on the model, independent of the statistical fluctuations of the data, and independent uncertainties add in variance. The Fisher matrix $F = D^T \hat{C}^{-1} D + P_h$ includes a Gaussian prior $P_h$ on the HOD parameters of width 0.29 times the prior range (the standard deviation of the uniform prior); the cosmological errors are marginalised over the HOD by inverting the full $18 \times 18$ matrix, and we also quote errors at fixed HOD. Marginalising over the HOD degrades the constraints by less than $\sim 30\%$, the most affected parameter being $n_s$: the HOD parameters partly mimic the effect of the cosmological parameters (e.g. a change of galaxy bias changes the amplitude of the clustering like $\sigma_8$), and leaving them free absorbs part of

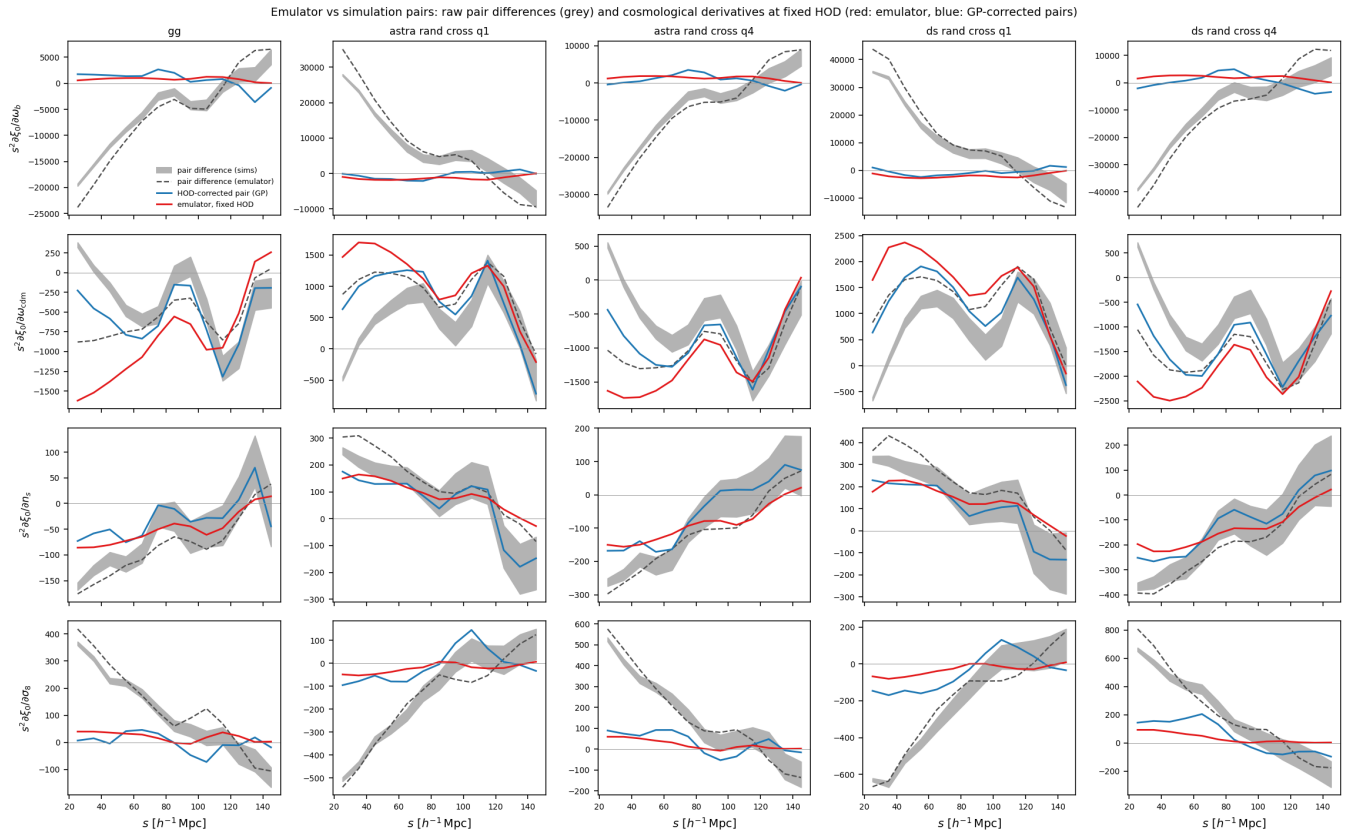



Figure 14: Cosmological derivatives at fixed HOD from the emulator (red) and from the pair differences corrected with the GP calibration of the HOD response (blue), with the raw pair differences in the simulations (grey) and as predicted by the emulator (dashed).

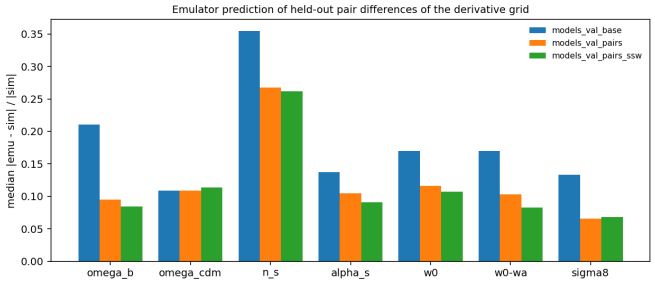



Figure 15: Median relative error of the emulator prediction of held-out pair differences of the derivative grid, for the three training variants (Table 4).

the cosmological signal; the small degradation shows that the statistics distinguish cosmology from galaxy formation well. The uncertainty due to the emulator training is estimated from the scatter of the forecasts of the individual networks, divided by $\sqrt{10}$.

**Compression.** Each data vector is compressed to one number per parameter (18 numbers), $\boldsymbol{t} = W^T \boldsymbol{d}$, $W = \tilde{C}^{-1} D$ ($\tilde{C}$: Ledoit–Wolf shrinkage estimate of the covariance), whose covariance is then estimated from the samples (Hartlap factor 0.90 with 192 samples, 0.99 with 1786). The compression is lossless if $\tilde{C} = C$; otherwise it keeps less information, which makes the forecast errors larger, never smaller, because the Fisher matrix of $\boldsymbol{t}$ is computed exactly with its own covariance (Appendix B); it allows long and combined data vectors to be compared fairly.

# 6 Analysis choices and how the results evolved

The framework above is the result of successive corrections, each motivated by a failure of the previous approach (Table 5, details in Appendix C). We report the intermediate results because they show which conclusions depend on the analysis choices.

**Results with the single-box covariance.** With the 192-subvolume covariance and the `hod239` fiducial, the baseline emulator and the regression over pairs, which rely on different assumptions, agreed that ASTRA split was as constraining as or slightly more constraining than density split for the random quartiles (Table 6): by 0–20% on $\omega_b$, $n_s$ and $\sigma_8$, the $\sigma_8$ ratio being above unity for both estimators. They disagreed on $\omega_{\rm cdm}$ (1.30 vs 0.94), and the absolute errors differed between the two estimators by a factor $\sim 2$ for $\omega_{\rm cdm}$, $n_s$, $\sigma_8$ and $\sim 6$ for $\omega_b$ (Table 7), as did the gain over the 2PCF: the regression

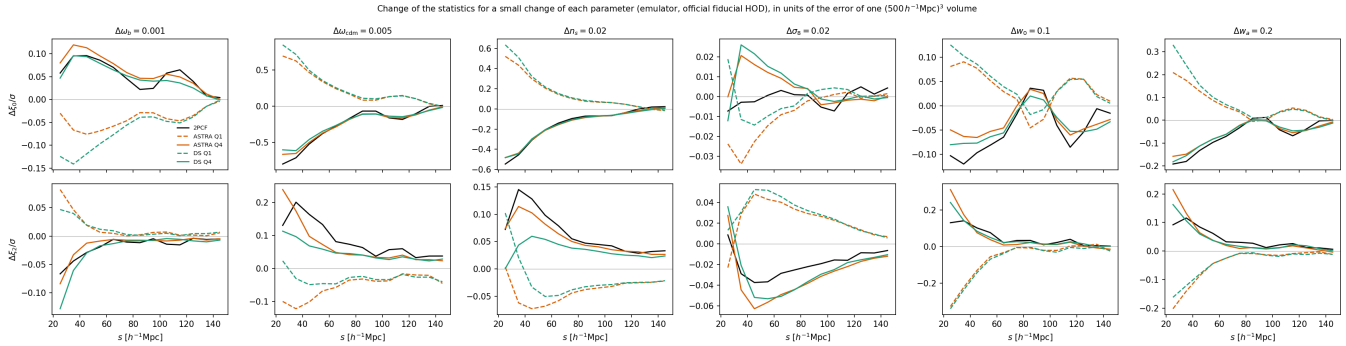



Figure 16: Change of the monopole (top) and quadrupole (bottom) for a small change of each parameter, in units of the error of one $(500\,h^{-1}\mathrm{Mpc})^3$ volume (baseline emulator at the official fiducial point): 2PCF (black), Q1 (dashed) and Q4 (solid) cross-correlations of ASTRA split (orange) and density split (green).

| Change | Problem solved | Effect on ASTRA vs DS |
|---|---|---|
| Averaged HOD files → HOD-matched pairs | the 500 files are different HOD models: their scatter ($\sim 55\%$) swamped the cosmology ($\sim 1\%$) | derivatives were pure noise |
| Finite differences → emulator | the residual HOD mismatch of the pairs dominated the derivatives by a factor $\sim 10$; a calibrated HOD correction (8% accurate) was not enough | Fisher matrices became well defined |
| Emulator → densified emulator trained on catalogue differences | $\omega_{\rm cdm}$ response not learnt; derivatives of individual networks scattered by 70–170%; ranking changed between trainings | ranking stable between trainings; cross-checked by an independent regression over the derivative grid |
| 192 subvolumes of one box → 1786 independent boxes | covariance from a single box, three correlated lines of sight | $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ on $\sigma_8$: 1.12 → 0.85–1.11; ASTRA's apparent lead disappears |
| DESI-like: 4 → 6 cosmological parameters ($+w_0$, $w_a$) | the emulator and derivative grid allow the dark-energy parameters of the initial goals | errors on $\omega_{\rm cdm}$ (+40%) and $n_s$ (+15%) degrade; ratios unchanged within errors |
| ASTRA classes → quartiles (main analysis), classes revisited with Quijote | with fixed thresholds, voids and knots contain $< 1\%$ of the points: two dominant classes | quartiles give balanced environments; classes + auto-correlations turn out the most constraining on Quijote |
| Quijote, 6 parameters → reliable subsets | derivative noise dominates the degenerate directions ($\Omega_b$, $h$, $M_\nu$) | ranking stable only for $(\Omega_m, \sigma_8)$, $(\Omega_m, n_s, \sigma_8)$ |

Table 5: The main changes of the analysis, the problem each solved, and its effect on the comparison.

is likely too optimistic (residual HOD contamination of its intercepts) and the emulator too conservative (training noise smoothing small slopes). The variants with auto-correlations or a single environment changed strongly between estimators and trainings.

**Results with the independent covariance.** With the same emulator derivatives, replacing the covariance by that of the 1786 independent boxes reversed the small lead of ASTRA split (Section 7). The 192 correlated samples misrepresented the relative noise of the two methods; the independent covariance is the more reliable one and is the baseline. What survived this change, and all the others, is that the two methods carry comparable information and are complementary.

**Baseline.** The baseline analysis was fixed after these steps: for the DESI-like mocks, the densified emulator trained on catalogue differences, the covariance of the official small boxes and the official HOD as fiducial point; for Quijote, the parameter subsets whose forecasts do not depend on the derivative noise. Other choices are treated as robustness tests.

# 7 Results

## 7.1 DESI-like mocks: ASTRA split, density split and the 2PCF

Table 8 and Figure 27 give the baseline forecasts. Both methods improve on the 2PCF by factors 1.6–3.1 depending on the parameter. Between them, the ratio $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ for the random-quartile cross-

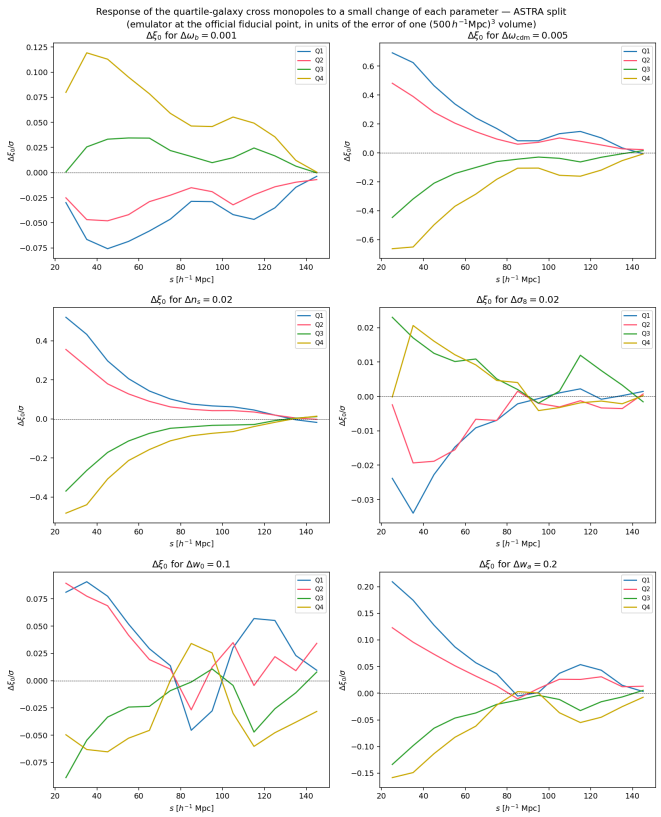



Figure 17: ASTRA split: change of the four quartile–galaxy cross monopoles for a small change $\Delta\theta$ of each parameter (value in each panel title), in units of the error of one $(500\,h^{-1}\mathrm{Mpc})^3$ volume (baseline emulator, official fiducial point).

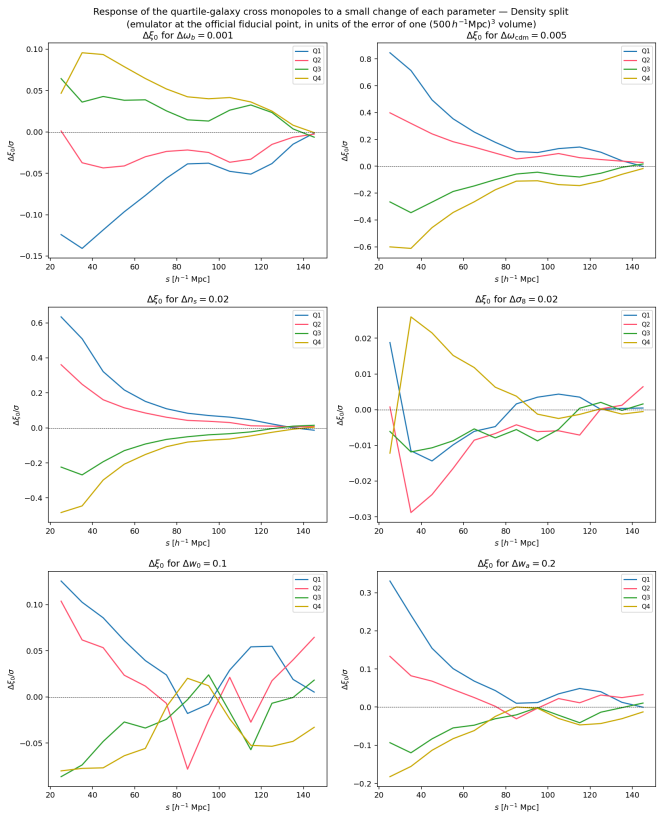



Figure 18: Same as Figure 17 for density split.

| quart. | deriv. | $\omega_b$ | $\omega_{\rm cdm}$ | $n_s$ | $\sigma_8$ |
|---|---|---|---|---|---|
| random | emulator | 1.02(15) | 1.30(11) | 1.11(9) | 1.12(6) |
| random | regression | 1.09(11) | 0.94(4) | 1.07(11) | 1.16(9) |
| galaxy | emulator | 0.80(9) | 0.91(8) | 0.92(7) | 0.96(2) |
| galaxy | regression | 1.08(16) | 1.15(10) | 1.10(13) | 1.00(12) |

Table 6: Superseded: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ with the 192-subvolume covariance (Q1+Q4 cross-correlations, HOD marginalised); above 1 favours ASTRA. In parentheses: uncertainty from the emulator ensemble or the bootstrap of the subvolumes.

| data vector | $\omega_b$ | $\omega_{\rm cdm}$ | $n_s$ | $\sigma_8$ |
|---|---|---|---|---|
| 2PCF | .023/.0026 | .018/.0067 | .20/.077 | .25/.083 |
| ASTRA | .011/.0018 | .015/.0071 | .11/.051 | .13/.081 |
| DS | .011/.0019 | .019/.0067 | .12/.054 | .14/.094 |
| A + DS | .0075/.0014 | .013/.0052 | .090/.039 | .11/.061 |
| all | .0066/.0011 | .010/.0042 | .081/.032 | .10/.052 |

Table 7: Superseded: marginalised errors with the 192-subvolume covariance, emulator / regression derivatives (random quartiles, compressed, HOD marginalised, one $(500\,h^{-1}\mathrm{Mpc})^3$ volume).

correlations is 0.90, 1.10, 0.84, 0.83, 0.94 and 0.92 for $\omega_b$, $\omega_{\rm cdm}$, $n_s$, $\sigma_8$, $w_0$ and $w_a$: density split is slightly ahead, by 5–15%, except on $\omega_{\rm cdm}$. The two are clearly complementary: combined, they improve on the better one by 23% ($\sigma_8$) to a factor 2.4 ($\omega_b$), 1.4 and 1.7 for $w_0$ and $w_a$, and the combination with the 2PCF is the most constraining (Figure 28). The errors on $w_0$ and $w_a$ ($\simeq 0.8$ and 2.7 for one $(500\,h^{-1}\mathrm{Mpc})^3$ volume) are larger than the range of the AbacusSummit suite: at this volume the dark-energy parameters are not constrained, and only the relative information content of the statistics is meaningful. Adding them nevertheless keeps the ASTRA errors on the other parameters moderate (at most +40%, on $\omega_{\rm cdm}$) and leaves the DS/ASTRA ratios stable, so the comparison does not depend on whether $w_0$ and $w_a$ are free. The variants (auto-correlations, galaxy quartiles, single environments) give ratios between 0.5 and 1.4 with no consistent winner (Figure 27).

Figure 19 shows the corresponding contours: the environment statistics mainly reduce the $\omega_b$–$\omega_{\rm cdm}$ and $n_s$–$\sigma_8$ degeneracies of the 2PCF.

**Individual quartiles.** Figures 20 and 21 show, as in the density-split analyses, the constraints of each quartile alone and of the four together, and Figures 22 and 23 compare the 2PCF, the void-like (Q1) and cluster-like (Q4) quartiles, all quartiles, and all quartiles with the 2PCF. A single quartile constrains the parameters about as well as the 2PCF, the inter-

| data vector | $\omega_b$ | $\omega_{\rm cdm}$ | $n_s$ | $\sigma_8$ | $w_0$ | $w_a$ |
|---|---|---|---|---|---|---|
| 2PCF | 0.033 | 0.057 | 0.40 | 0.58 | 2.1 | 6.7 |
| ASTRA | 0.012 | 0.032 | 0.22 | 0.23 | 0.80 | 2.7 |
| DS | 0.011 | 0.035 | 0.18 | 0.19 | 0.75 | 2.5 |
| ASTRA + DS | 0.0046 | 0.022 | 0.13 | 0.15 | 0.53 | 1.5 |
| 2PCF + ASTRA + DS | 0.0039 | 0.019 | 0.12 | 0.13 | 0.50 | 1.3 |

Table 8: Baseline marginalised $1\sigma$ errors (random-quartile Q1+Q4 cross-correlations, HOD marginalised) for one $(500\,h^{-1}{\rm Mpc})^3$ volume. These values are comparable to the prior widths and compare information content; they are not realistic constraints.

| fiducial HOD | $\omega_b$ | $\omega_{\rm cdm}$ | $n_s$ | $\sigma_8$ | $w_0$ |
|---|---|---|---|---|---|
| official | 0.90(11) | 1.10(16) | 0.84(15) | 0.83(10) | 0.94(1 |
| hod239 | 0.83(9) | 0.92(7) | 1.04(12) | 0.93(8) | 0.88(6 |
| hod212 | 0.77(7) | 0.94(9) | 0.94(13) | 0.84(8) | 0.85(8 |
| hod795 | 0.79(8) | 0.98(10) | 1.06(16) | 1.14(10) | 0.89(1 |

Table 9: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ (above 1: ASTRA ahead) for the random-quartile cross-correlations with the official covariance at four fiducial HODs; in parentheses, the uncertainty from the emulator training.

mediate quartiles (Q3 in particular) being the least constraining, but its information is of a different nature: the 2PCF averages the clustering over all environments, whereas a quartile measures it around one range of densities only, e.g. the profile of galaxies around voids for Q1. Each environment responds to the parameters in its own way: the void and cluster quartiles respond with opposite signs and different scale dependences (Figures 17, 18), and their degeneracy directions differ. The gain therefore comes from combining environments: the four quartiles together are several times tighter than any one of them or than the 2PCF, which mixes all environments and loses this distinction. Figure 24 summarises the information per environment: the Q4 cross-correlations alone already beat the 2PCF on most parameters, Q1 and Q4 together double the gain, and adding the auto-correlations multiplies it again.

**Auto-correlations.** Adding the quartile auto-correlations to the cross-correlations tightens the constraints of both methods by factors 1.3–2.8 depending on the parameter (Figure 25); with them, the two methods remain comparable (ratios 0.7–1.1), density split being ahead on $\omega_b$, $\omega_{\rm cdm}$ and $n_s$ and ASTRA on $\sigma_8$.

**Scale cuts.** Figure 26 shows the errors as a function of the minimum scale. All statistics lose information as small scales are removed, the environment statistics staying 0.2–0.5 dex below the 2PCF up to $s_{\rm min} = 70\,h^{-1}$Mpc (beyond, the 2PCF has fewer bins than parameters). ASTRA and density split stay within 0.1 dex of each other at all scale cuts, their ranking changing with $s_{\rm min}$, and their combination keeps its 0.2–0.4 dex advantage.

**Fiducial point.** Evaluated at four different HODs, the ratio $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ stays between 0.77 and 1.14 for all six parameters (Table 9; 0.75–1.11 with four parameters): the fiducial point matters at the level of $\pm 15\%$.

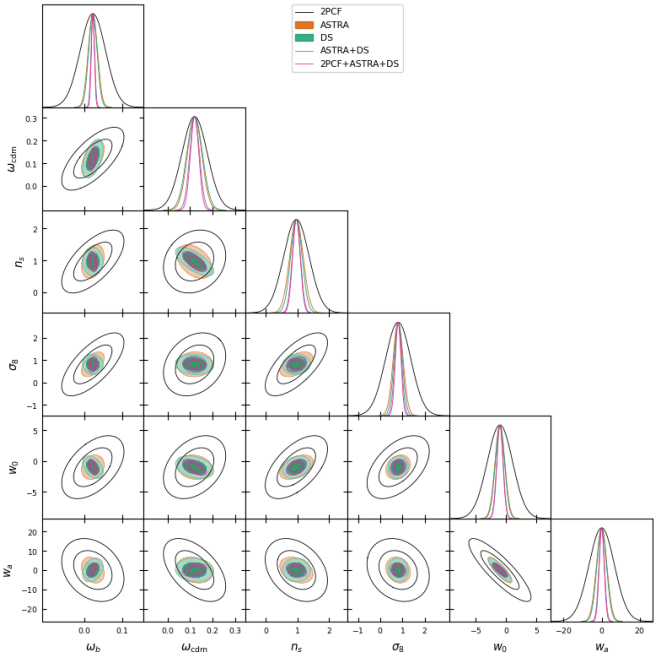



Figure 19: Baseline Fisher contours (68% and 95%) for the 2PCF, ASTRA split, density split and their combinations, one $(500\,h^{-1}{\rm Mpc})^3$ volume, HOD marginalised.

## 7.2 Quijote halos

**Covariance.** With 4000 fiducial realisations the covariance is converged: at fixed derivatives the marginalised errors change by less than 1% between 2000 and 4000 realisations, and by less than 6% between 1000 and 4000. (About half of the 15 517 fiducial catalogues of our copy were empty, an incomplete download without consequence given this convergence; all derivative sets are complete.)

**Pairing.** The $\pm$ realisations of a given index share their initial conditions: the scatter of the differences between realisations of the same index is 3–4 times smaller than between different indices, for the 2PCF as for the quartile statistics.

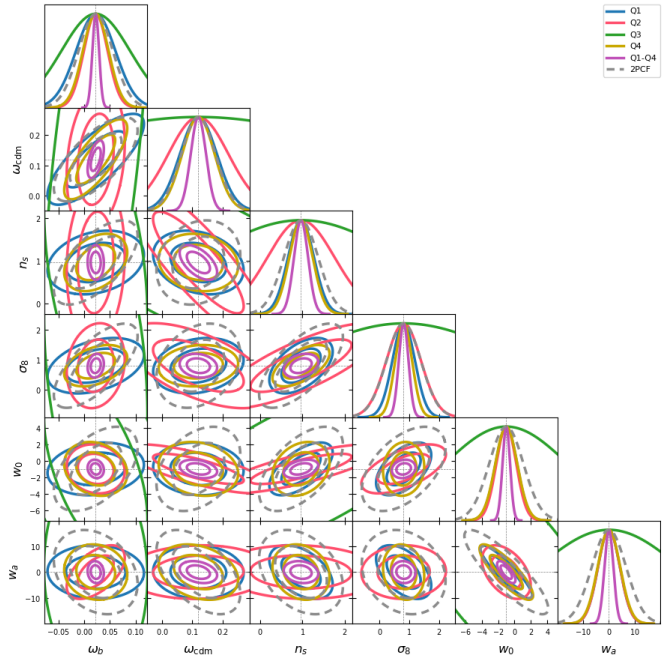



Figure 20: Constraints (68% and 95%) from individual and combined ASTRA-split quartiles (cross-correlations) and from the 2PCF (dashed), baseline, HOD marginalised. Axes span $\pm 3\sigma$ of the 2PCF.

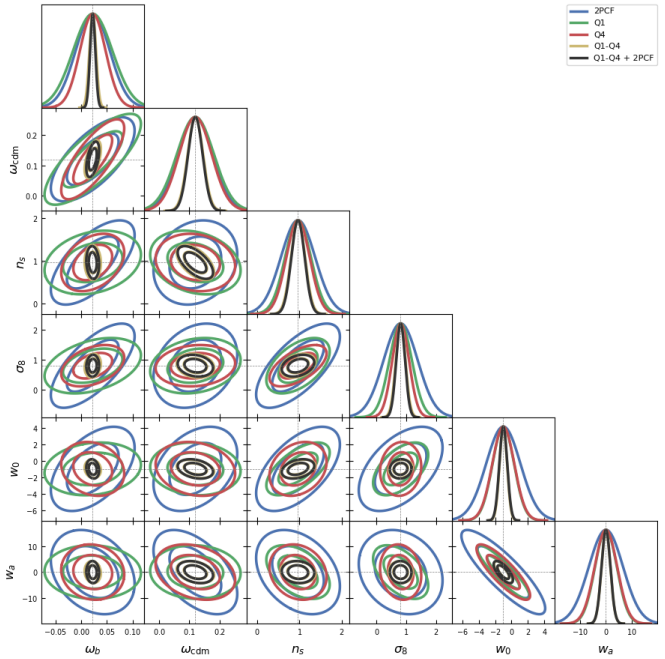



Figure 22: ASTRA split: constraints from the 2PCF, the Q1 and Q4 cross-correlations, all quartiles, and all quartiles with the 2PCF (baseline).

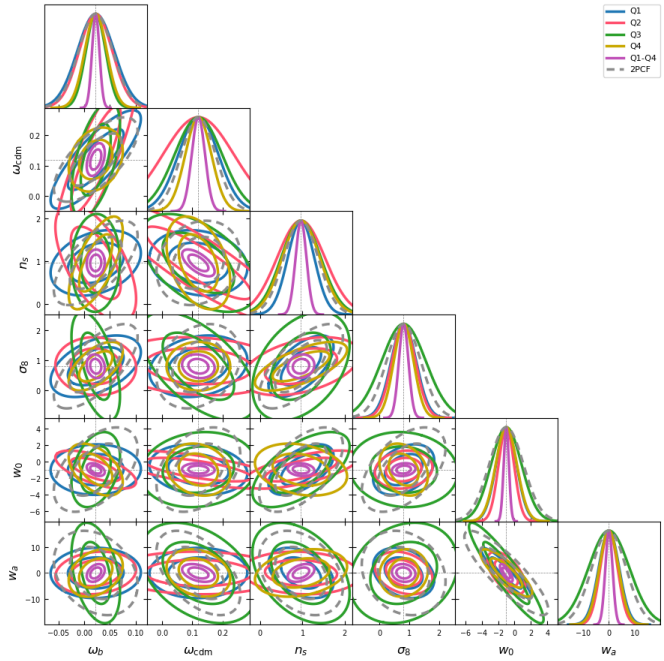



Figure 21: Same as Figure 20 for density split.

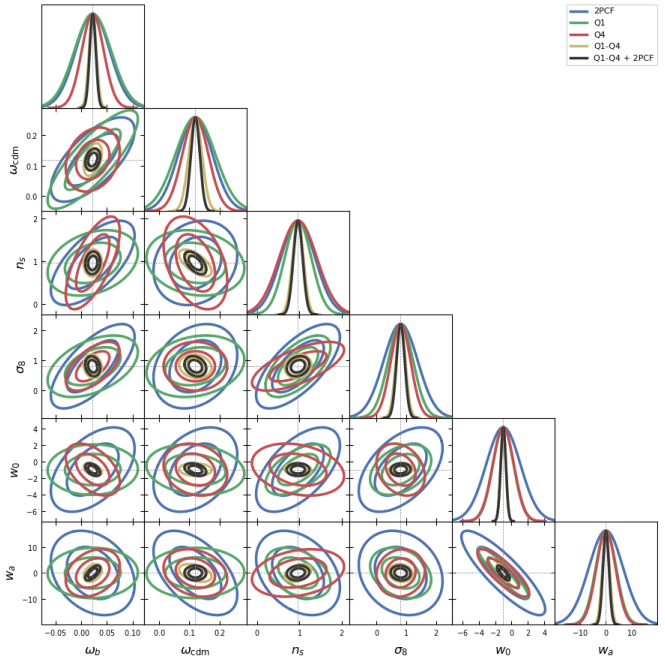



Figure 23: Same as Figure 22 for density split.

**Noise of the derivatives.** Even paired, the derivatives remain noisy. Quijote is designed for statistics of the matter and halo fields (power spectrum, bispectrum, ...). Our observables add two sources of noise that the pairing cannot cancel:

1. *Sparse tracers.* With $\simeq 50\,000$ host halos per $(h^{-1}\mathrm{Gpc})^3$, each quartile contains only $\simeq 12\,000$ halos (or random points), and the shot noise of the quartile statistics is large.

2. *Discrete classification.* A small change of cosmology moves halos and random points from one quartile to another. This re-classification is only partly common to the + and − simulations, even with identical initial conditions, and adds a noise that does not cancel in the difference.

This noise is mostly amplified by the six-parameter Fisher matrix. $\Omega_b$, $h$ and $n_s$ (and $M_\nu$) are strongly degenerate in the clustering of halos: their weakest combination is constrained by a very small amount

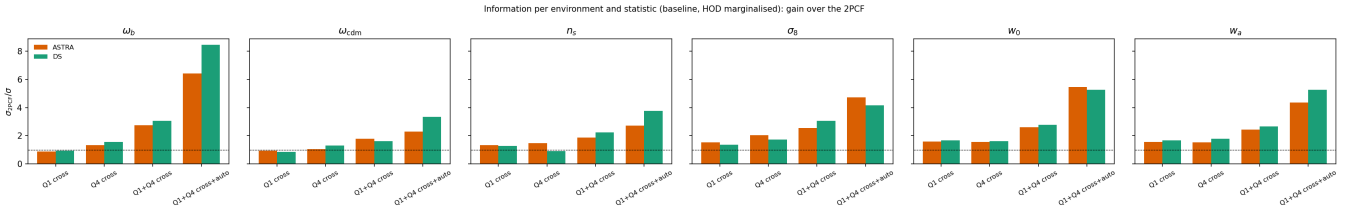



Figure 24: Gain over the 2PCF, $\sigma_{\rm 2PCF}/\sigma$, for the Q1 and Q4 cross-correlations alone, together, and with the quartile auto-correlations, for ASTRA split (orange) and density split (green) (baseline, HOD marginalised).

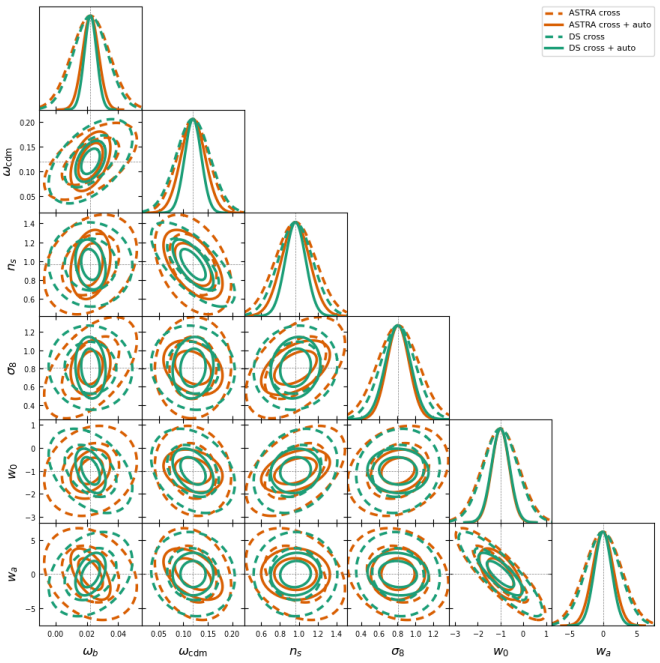



Figure 25: Q1+Q4 cross-correlations only (dashed) and with the quartile auto-correlations (solid), for ASTRA split (orange) and density split (green), baseline.

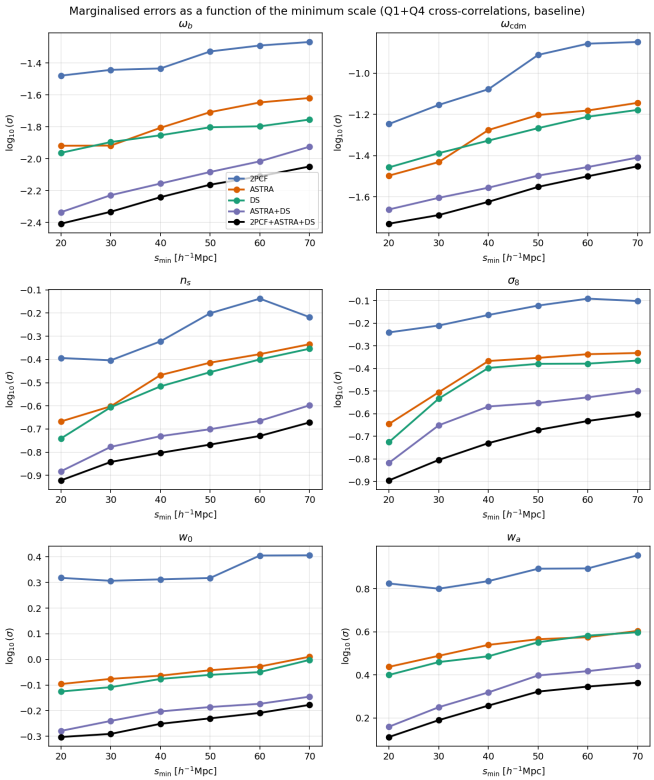



Figure 26: Marginalised errors as a function of the minimum scale $s_{\rm min}$ (Q1+Q4 cross-correlations, baseline).

of information, of the same order as the noise left in the average of 500 derivative pairs, so that along this direction the derivatives are dominated by noise. The standard Fisher estimator is then biased high, and subtracting the expected noise term makes the six-parameter matrices non positive definite (Appendix B).

**Estimator and reliability criterion.** We therefore use the approach of [13]: the standard estimator is biased high by the derivative noise, a compressed estimator (half of the simulations define the compression, the other half measure its information) is biased low; the two bracket the true Fisher information, and their geometric mean is nearly unbiased. Where they agree, the forecast does not depend on the derivative noise: we call a forecast *reliable* when the compressed and standard errors differ by less than a factor 1.5 for both statistics compared.

**Choice of the parameters.** The bias grows with the length of the data vector and is concentrated in the degenerate directions. We computed the forecasts for six parameter subsets, from $(\Omega_m, \sigma_8)$ to all six parameters, and three compressions of the data vector ($10\,h^{-1}$Mpc bins with $\ell = 0, 2$; $20\,h^{-1}$Mpc bins with $\ell = 0, 2$; $20\,h^{-1}$Mpc bins, monopole only; Figure 29). Only $(\Omega_m, \sigma_8)$ and $(\Omega_m, n_s, \sigma_8)$ give reliable forecasts, for all compressions; as soon as $\Omega_b$, $h$ or $M_\nu$ are varied, the two estimators differ by factors 2–10 and the ranking of the methods changes from one compression to the next. These parameters cannot be constrained with these observables and 500 derivative pairs. Fixing $\Omega_b$, $h$ (and $n_s$), as many density-split forecasts do, is justified by external constraints (BBN, CMB); $M_\nu$ would require denser tracers or many more derivative simulations.

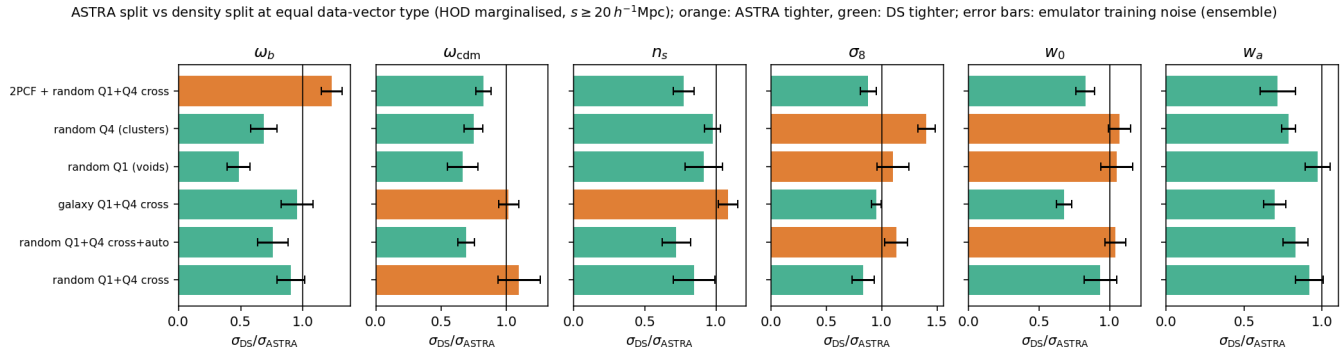



Figure 27: Baseline: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ for several data-vector types (above 1: ASTRA split more constraining). Error bars: uncertainty from the emulator training.

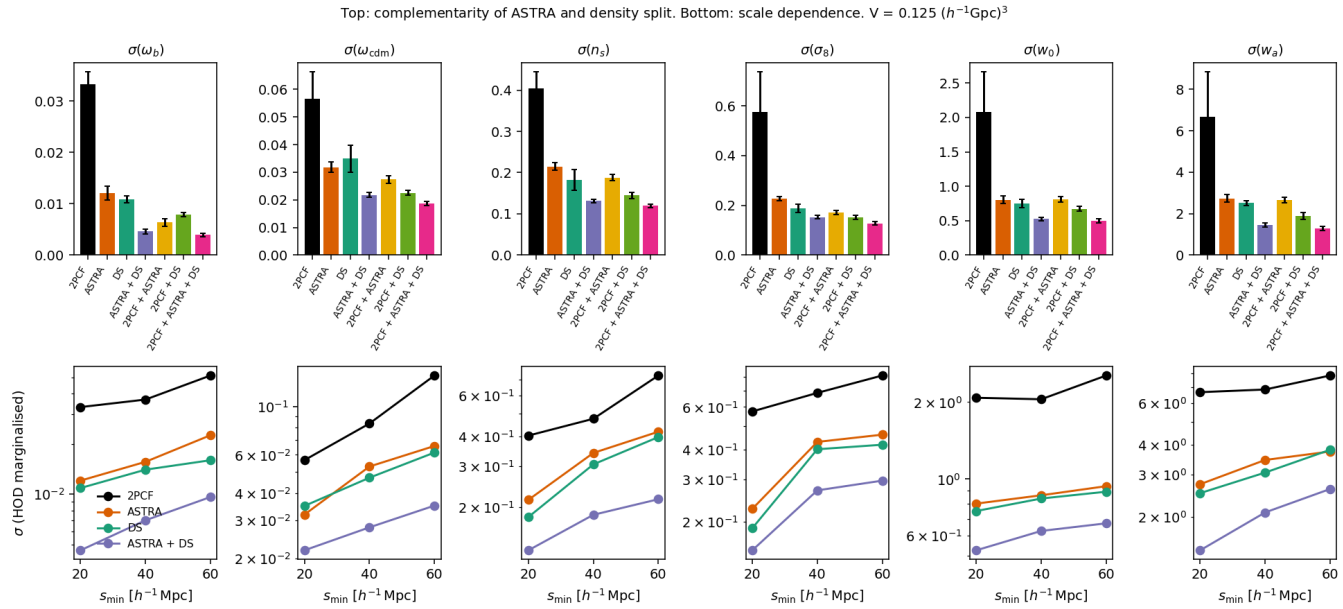



Figure 28: Baseline. Top: marginalised errors of the 2PCF, ASTRA split, density split and their combinations. Bottom: dependence on the minimum scale.

**Results.** The reliable forecasts (Table 10) agree with the DESI-like analysis for the random-quartile cross-correlations: density split is as constraining as ASTRA split or more ($\sigma_{\rm DS}/\sigma_{\rm ASTRA} = 0.6$–$1.1$). When the quartile auto-correlations are added, or with the halo quartiles, ASTRA split is more constraining on $\sigma_8$ by a factor 1.5–2 and comparable on $\Omega_m$ and $n_s$: its information lies more in the clustering of its environments with themselves and in the environments of the tracers than in the tracer profiles around random environments.

| quartiles | parameters | $\Omega_m$ | $n_s$ | $\sigma_8$ |
|---|---|---|---|---|
| random, cross | $\Omega_m, \sigma_8$ | 0.86–1.04 | – | 0.65–0.81 |
| random, cross | $\Omega_m, n_s, \sigma_8$ | 0.57–0.64 | 0.68–0.74 | 1.04–1.14 |
| random, + auto | $\Omega_m, \sigma_8$ | 1.19–1.40 | – | 1.54–2.11 |
| random, + auto | $\Omega_m, n_s, \sigma_8$ | 0.74–0.81 | 0.85–0.92 | 1.72–1.78 |
| halo, cross | $\Omega_m, \sigma_8$ | 1.18–1.21 | – | 1.91–2.04 |
| halo, cross | $\Omega_m, n_s, \sigma_8$ | 1.16–1.20 | 1.17–1.23 | 1.54–1.70 |

Table 10: Quijote, reliable forecasts: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$, range over three compressions of the data vector (above 1: ASTRA ahead).

## 7.3 ASTRA cosmic-web classes (Quijote)

To test whether the original ASTRA classes carry more information than quartiles, we measured them on Quijote, where the derivatives are clean: 4000 fiducial realisations for the covariance and the 500 pairs of $\Omega_m$, $n_s$ and $\sigma_8$, with the same halos, random points and estimators as the quartile analysis, and compared them with the quartiles of ASTRA and density split measured on the same realisations ($10\,h^{-1}$Mpc bins, $\ell = 0, 2$, $s \geq 20\,h^{-1}$Mpc). Because the void and knot classes are nearly empty for random points, the class data vectors use the sheets and filaments (adding the voids changes nothing).

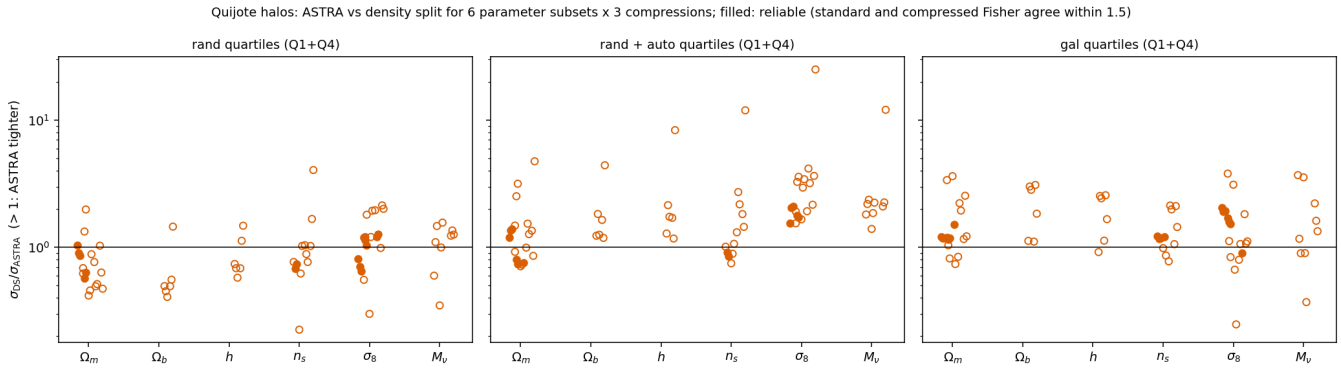



Figure 29: Quijote halos: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ (above 1: ASTRA more constraining) for six parameter subsets and three compressions, for the random-quartile cross-correlations (left), with their auto-correlations (centre), and the halo quartiles (right). Filled: reliable forecasts (standard and compressed Fisher estimators within a factor 1.5); open: dominated by the noise of the derivatives.

**Class fractions.** The fractions of points in each class are themselves sensitive to cosmology (Figure 30): a variation of $\Omega_m$, $n_s$ or $\sigma_8$ shifts the sheet and filament fractions of the random points by 0.3–1 times their scatter between realisations, in opposite directions. Unlike quartiles, which have fixed fractions by construction, fixed thresholds let the volume of each environment respond to cosmology.

**Information.** Table 11 and Figure 31 compare the classes with the quartiles of ASTRA and density split for the three compressions of the data vector used in Section 7.2. For $(\Omega_m, \sigma_8)$ all the forecasts are reliable. With cross-correlations only, the classes carry about as much information as the ASTRA quartiles, and less than density split on $\sigma_8$. With the auto-correlations, the classes are the most constraining statistic for every compression: $\sigma(\sigma_8)$ is 2.1–3.0 times smaller than with density split and 1.35–1.45 times smaller than with the ASTRA quartiles, and $\sigma(\Omega_m)$ 1.3–1.5 and $\sim 1.1$ times smaller. For $(\Omega_m, n_s, \sigma_8)$, the classes with auto-correlations are reliable for all compressions (and are the only reliable ASTRA random-point vector with the finest binning); their advantage over density split remains a factor 2.8–2.9 on $\sigma_8$, but shrinks with the compression on $\Omega_m$ (1.0–1.5) and $n_s$ (1.2–1.85). The classes of the halos give $\sigma(\Omega_m)$ 15% smaller than the halo quartiles but a larger $\sigma(\sigma_8)$.

**Interpretation.** The gain comes from the auto-correlations of the two classes and from the response of their volumes to cosmology, which equal-size quartiles remove by construction. The advantage on $\sigma_8$, a factor 2–3 over density split, is robust to the compression, the parameter set and the number of covariance realisations; the advantage on $\Omega_m$ and $n_s$ is smaller and depends on the compression. For a DESI-like analysis, this would require an emulator trained on class statistics, i.e. remeasuring the emulator suite.

| compression | $\Omega_m$ | $n_s$ | $\sigma_8$ |
|---|---|---|---|
| $(\Omega_m, \sigma_8)$ | | | |
| $10\,h^{-1}$Mpc, $\ell = 0, 2$ | 1.28 | – | 2.08 |
| $20\,h^{-1}$Mpc, $\ell = 0, 2$ | 1.49 | – | 2.95 |
| $20\,h^{-1}$Mpc, $\ell = 0$ | 1.52 | – | 2.96 |
| $(\Omega_m, n_s, \sigma_8)$ | | | |
| $10\,h^{-1}$Mpc, $\ell = 0, 2$ | 1.51* | 1.85* | 2.88* |
| $20\,h^{-1}$Mpc, $\ell = 0, 2$ | 1.07 | 1.31 | 2.86 |
| $20\,h^{-1}$Mpc, $\ell = 0$ | 0.99 | 1.20 | 2.78 |

Table 11: Quijote, 4000 fiducial realisations: $\sigma_{\rm DS}/\sigma_{\rm classes}$ for the random-point cross- and auto-correlations of the density-split quartiles and of the ASTRA classes (sheets and filaments); above 1: classes more constraining. *: density split marginally reliable (estimator ratio 1.5–1.6).

## 7.4 Inference with the emulator (MCMC)

Fisher forecasts assume Gaussian posteriors and are local. To test the full pipeline as it would be applied to data, we sampled the posterior of the 20 parameters (8 cosmological, 12 HOD) with the emulator as the model: Gaussian likelihood with the official covariance rescaled to the data volume plus the emulator error, uniform priors over the training range, `emcee` with 64 walkers and 8000 steps (Appendix B). The data are measurements the emulator has never seen: one subvolume of each of 2 catalogues of the 8 test cosmologies, and 8 subvolumes of the official base box of the fiducial cosmology, each analysed with the 2PCF, ASTRA split, density split and their combinations (120 chains).

The posteriors are unbiased and calibrated or slightly conservative: the truth lies within $1\sigma$ of the posterior mean in 70–100% of the runs depending on the parameter (68% expected). They are however barely narrower than the priors (Figures 32 and 33): one $(500\,h^{-1}\text{Mpc})^3$ volume with 12 free HOD param-

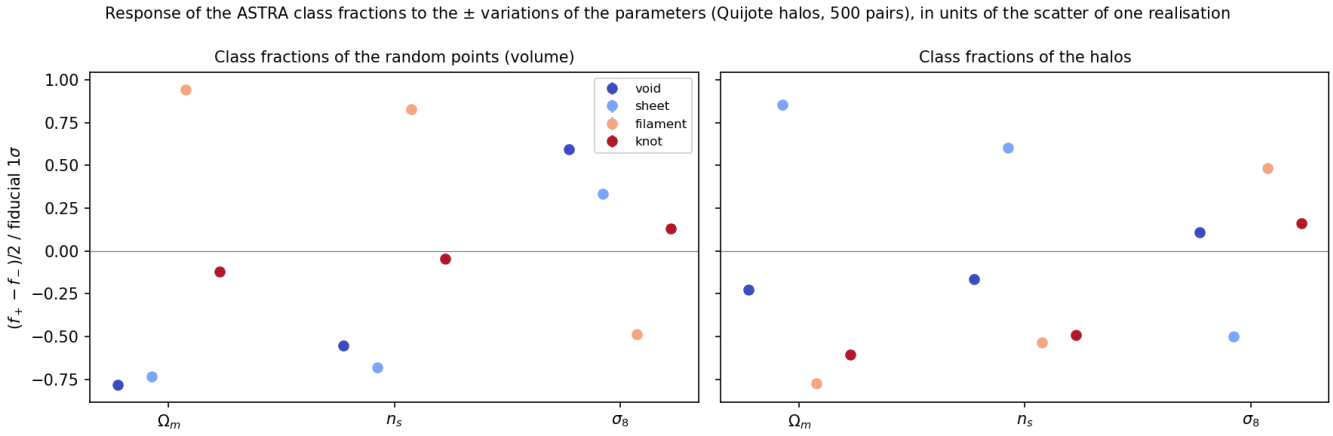



Figure 30: Response of the ASTRA class fractions of the random points (left) and halos (right) to the ± variations of $\Omega_m$, $n_s$ and $\sigma_8$ (Quijote, 500 pairs), in units of the scatter of one realisation (error bars: error of the mean, smaller than the symbols).

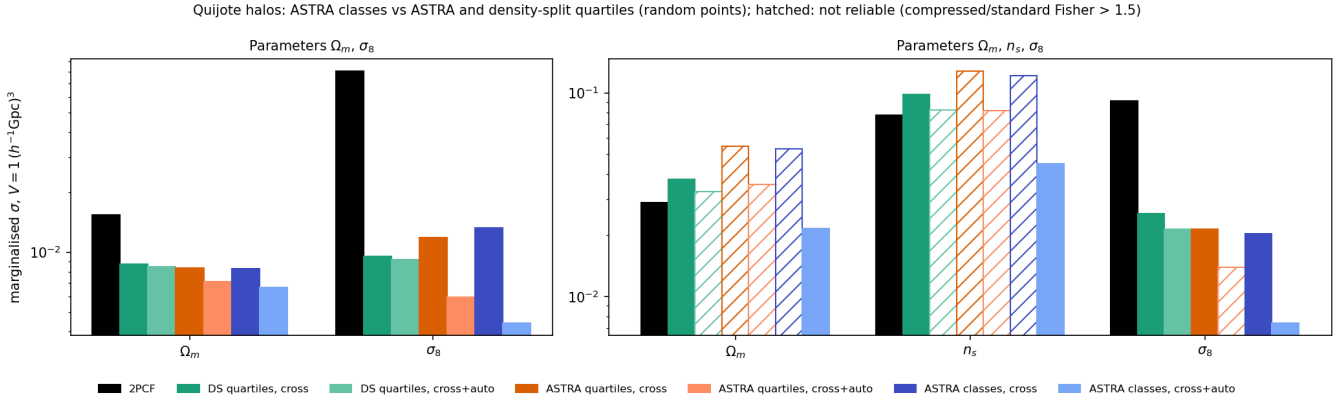



Figure 31: Quijote halos, $V = 1\,(h^{-1}\mathrm{Gpc})^3$: marginalised errors for $(\Omega_m, \sigma_8)$ (left) and $(\Omega_m, n_s, \sigma_8)$ (right) from the 2PCF, the density-split and ASTRA quartiles of the random points, and the ASTRA classes (sheets and filaments), with cross-correlations only or with auto-correlations. Hatched: not reliable (compressed and standard Fisher estimators differ by more than a factor 1.5).

eters constrains only $\omega_{\mathrm{cdm}}$ and $\sigma_8$ beyond the prior, by 10–15%, and the statistics differ by a few per cent. A first set of runs on the full $(2\,h^{-1}\mathrm{Gpc})^3$ box was limited the other way: the emulator error (0.4–0.9 times the statistical error of that volume) was as large as the statistical error, and masked the differences between the statistics. Distinguishing the statistics through inference therefore requires both a large volume and an emulator error well below the statistical error of that volume; with the present training set this is not the case (and the learning curve shows that more catalogues of the same kind would not change it), so the comparison of the methods rests on the Fisher forecasts. The inference pipeline is validated and ready for a larger or less noisy training set.

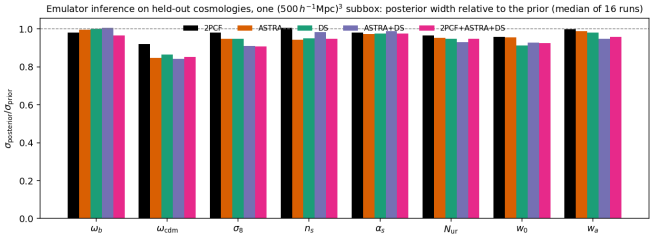



Figure 32: Emulator inference on held-out cosmologies (one $(500\,h^{-1}\mathrm{Mpc})^3$ volume, official covariance): posterior width relative to the prior width, median of 16 runs, for the 2PCF, ASTRA split, density split and their combinations.

## 8 Robustness tests

We test the sensitivity of the measurements to the numerical choices on the first 8 subvolumes of the fiducial catalogue (line of sight $z$). Each test is repeated for both methods, and the change of the Q1 and Q4 cross-correlation monopoles is expressed in units of the statistical error of one subvolume (Figure 34).

**Periodic padding.** Increasing the padding around each subvolume from 30 to 60 $h^{-1}$Mpc leaves the AS-

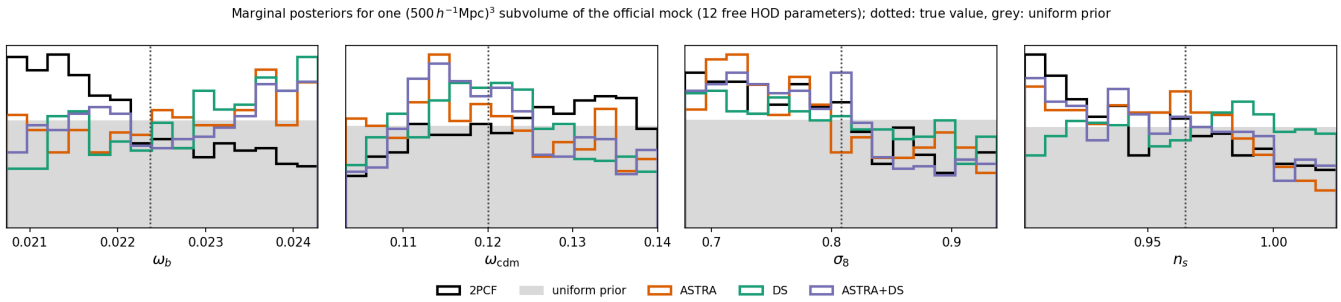



Figure 33: Marginal posteriors of four cosmological parameters for one $(500\,h^{-1}\mathrm{Mpc})^3$ subvolume of the official mock (fiducial cosmology, dotted lines), with 12 free HOD parameters, for the 2PCF, ASTRA split, density split and ASTRA + DS. At this volume all posteriors remain close to the uniform prior (grey).

TRA values of $r$, and therefore all the statistics, unchanged: the Delaunay neighbourhood of every point of the subvolume is already contained in the $30\,h^{-1}$Mpc padding.

**Number of randoms.** Doubling the number of random points in the tessellation ($N_r = 2N_g$) shifts the ASTRA Q1 cross-correlations by up to $4\sigma$ and the Q4 ones by up to $2\sigma$ at $s \lesssim 40\,h^{-1}$Mpc, while density split, for which the randoms are only query points, changes by less than $0.2\sigma$. In ASTRA the randoms enter the definition of $r$ itself (Eq. 1), so $N_r/N_g$ is a parameter of the statistic and must be kept fixed.

**Sampling density.** Randomly thinning the galaxies (and the randoms in the same proportion) by a factor 2 (5) shifts the ASTRA cross-correlations by 7–10 (20–35)$\sigma$ at $s \simeq 20$–$40\,h^{-1}$Mpc, against $\sim 1$ (3–5)$\sigma$ for density split. At the level of individual points, the rank correlation between the $r$ of the same galaxies computed in the full and in the thinned samples is only 0.65, 0.38 and 0.24 for thinning factors of 2, 5 and 10, and only 58%, 42% and 36% of the galaxies of the extreme quartiles keep their quartile.

This is an intrinsic property of the estimator rather than a numerical instability. ASTRA is an *adaptive-scale* statistic: the environment of a point is defined by its Delaunay neighbours, so its effective scale is the typical distance between neighbours, $\propto \bar{n}^{-1/3}$, and it changes with the sampling density. Thinning enlarges this scale, which is why the changes are concentrated at $s \lesssim 50\,h^{-1}$Mpc, whereas density split keeps a fixed smoothing radius. This is not a limitation for the present analysis or for DESI, since the density is fixed in the mocks as in the data (and in an emulator trained at that density). It implies however that ASTRA statistics must be compared, modelled and emulated at the same number density and random-to-galaxy ratio, and that comparisons with density split must be made at equal density, as throughout this paper.

# 9 Discussion

**What is robust.** The comparison was made with three derivative estimators (emulator, regression over pairs, Quijote finite differences), two covariances, four fiducial HODs, two tracer populations (LRG-like galaxies, halos) and several data vectors. Two conclusions never changed: ASTRA split and density split carry comparable information (error ratio 0.75–1.2 for the same data vector, with four or six parameters, in the DESI-like analysis with the independent covariance, 0.6–2 in the reliable Quijote forecasts), and they are complementary, their combination gaining 23% to a factor 2.4. The ranking for individual parameters changed with the analysis choices (Table 5), at the 10–40% level, which is the size of the effect of those choices; we therefore do not claim a winner. Where a trend is consistent, density split is slightly ahead on $\omega_b$, $\Omega_m$ and $n_s$ with the random-quartile cross-correlations, and ASTRA split ahead on $\sigma_8$ through its environment auto-correlations and tracer environments (Quijote). In particular, the advantage of ASTRA split on $\sigma_8$ found with the single-box covariance disappears with the independent covariance.

**What we learnt about the methodology.** (i) Emulator suites in which the HOD is drawn independently in each cosmology do not allow naive finite differences: the HOD mismatch dominates the cosmological response by an order of magnitude, and paired differences do not cancel the galaxy sampling noise of different HOD catalogues. (ii) Single-box covariances, even with many subvolumes and lines of sight, can bias the *comparison* of statistics, not only their absolute errors. (iii) Environment statistics of sparse tracers have noisy derivatives, which a six-parameter Fisher matrix amplifies along its degenerate directions; comparing two Fisher estimators that bracket the truth identifies the reliable forecasts. (iv) The clustering statistics respond essentially to a single combination of the 12 HOD parameters.

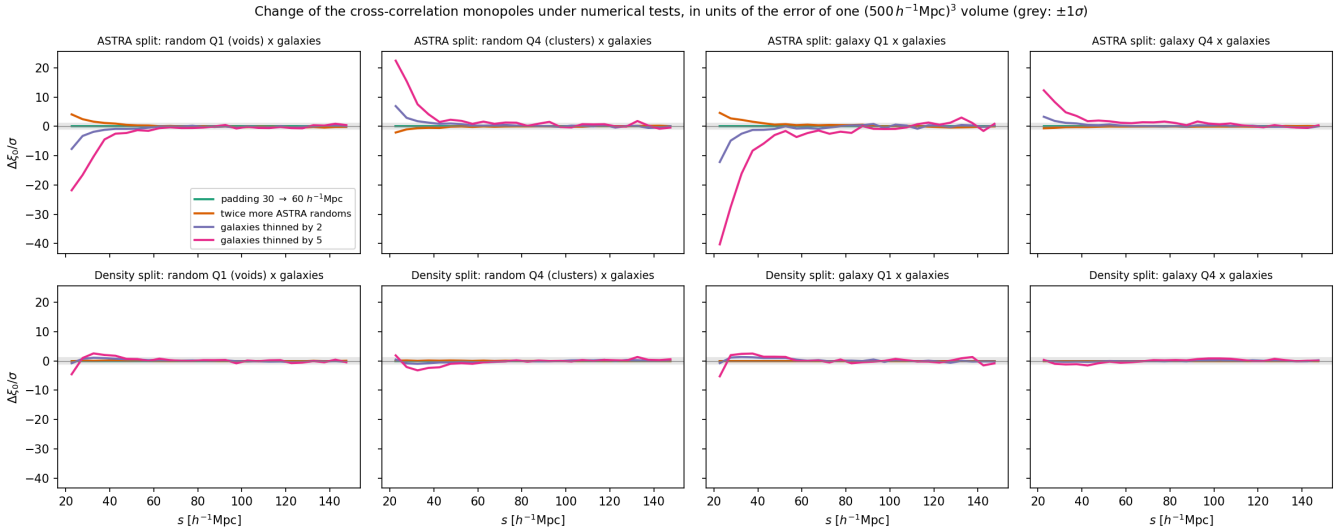



Figure 34: Change of the Q1 and Q4 cross-correlation monopoles (random quartiles, left; galaxy quartiles, right) of ASTRA split (top) and density split (bottom) when the padding is doubled (green), the ASTRA randoms are doubled (orange), and the galaxies are thinned by 2 (purple) and 5 (pink), relative to the baseline, in units of the error of one $(500\,h^{-1}\mathrm{Mpc})^3$ volume (grey band: $\pm1\sigma$). Mean over 8 subvolumes of the fiducial catalogue.

**Limitations.** (i) The forecasts are for one $(500\,h^{-1}\mathrm{Mpc})^3$ volume, where the errors are comparable to the prior widths: they compare information content and are not realistic constraints. (ii) The inference is calibrated but not yet discriminating: at this volume the prior dominates, and for a full box the emulator error does. (iii) The Quijote halos are ten times sparser than the LRGs, and ASTRA's adaptive scale is correspondingly larger. (iv) The emulator suite has a single HOD realisation per point, whose galaxy sampling noise sets the floor of the emulator error.

**Improvements.** A learning curve shows that the emulator error has reached a plateau at about half the statistical error of a $2\,h^{-1}$Gpc box: more training catalogues of the same kind would barely help. Several HOD realisations (seeds) per point, or HOD-matched catalogues generated with AbacusHOD in all cosmologies, are needed both to lower this floor, which would make an inference on a DESI-like volume discriminating, and to obtain clean derivatives in the DESI-like setting. Larger sets of independent boxes would remove the remaining dependence on the covariance.

**Beyond quartiles.** Being local and free of smoothing, ASTRA provides more than a density ranking: a classification of every galaxy into voids, sheets, filaments and knots, a continuous environment value usable as a mark, and a natural treatment of survey masks through the random points. The first test of the classes on Quijote (Section 7.3) is the most promising result of this work for ASTRA: despite their imbalance, the classes with their auto-correlations constrain $\sigma_8$ two to three times better than density split, robustly across compressions. Natural next steps are the probabilistic classes of [4] (classification repeated over many random catalogues, which gives every point a probability of belonging to each class and could make the void and knot classes usable), thresholds adapted to the density of the sample, and an emulator of class statistics for DESI.

# 10 Conclusions

We set out to measure how much cosmological information ASTRA environment statistics add to the galaxy two-point function, and how they compare with density split. We measured both on the same DESI-like mocks and Quijote halos, built covariance and derivative datasets for Fisher forecasts, and tested the full chain with an emulator-based inference. The main points are:

- **Gain over the two-point function.** ASTRA environment statistics constrain $\omega_b$, $\omega_{\mathrm{cdm}}$, $n_s$, $\sigma_8$, $w_0$ and $w_a$ 1.6–3.1 times better than the galaxy two-point function.
- **Comparison with density split.** The two methods carry comparable information (error ratio 0.8–1.1, density split slightly ahead) and are complementary: combining them gains another 23% to a factor 2.4. With Quijote halos, ASTRA does better on $\sigma_8$ (factor 1.5–2) through

its environment auto-correlations and tracer environments.

- **Cosmic-web classes.** With the fixed ASTRA thresholds, the classes reduce in practice to two dominant classes (sheets and filaments) at these densities, which is why the main analysis uses quartiles. On Quijote, the classes with their auto-correlations are nevertheless the most constraining statistic: two to three times better than density split on $\sigma_8$, robustly across compressions.
- **What ASTRA measures.** The ASTRA and density-split environments are strongly but not perfectly correlated ($\rho \simeq 0.72$). The ASTRA value $r$ itself hardly responds to cosmology: the information lies in how the environments cluster. ASTRA adapts its scale to the local density, so it must be modelled at the density and random-to-galaxy ratio of the data.
- **Method.** Emulator suites with an independent HOD per cosmology do not allow naive finite differences, single-box covariances can bias the comparison of statistics, and environment statistics of sparse tracers have noisy derivatives. We showed how each problem was diagnosed and addressed. The emulator inference is unbiased and calibrated, but discriminating between statistics with it would require several HOD realisations per training point.

Natural continuations, beyond the scope of this paper, are the probabilistic ASTRA classes of [4], which could make the void and knot classes usable, and an emulator of the class statistics for a DESI-like analysis of the classes.

# A Reproducing this paper

**Code.** All the measurements, forecasts and figures are produced by the code at `https://github.com/Hugoooat/DESI_astra/tree/astra_hugo_v2`, run on NERSC Perlmutter CPU nodes with the DESI `cosmodesi` environment:

```
source /global/common/software/desi/users/
    adematti/cosmodesi_environment.sh main
```

The code is organised in three modules and a set of scripts: `astra_ds.py` contains the whole measurement of one catalogue (reading, periodic padding, ASTRA tessellation, density field, quartiles, subvolumes, clustering), `analysis.py` the loading of measurements, data vectors, covariances and Fisher tools, `emulator.py` the emulator (networks, training with the difference loss, derivatives). Each catalogue is measured on one 128-core node in $\simeq 8$ minutes (64 subvolumes processed in parallel, ASTRA and density split, 33 statistics, three multipoles); measurements are saved as one `.npz` file per catalogue, line of sight and space, containing the multipoles of every statistic for every subvolume. Lists of catalogues are run with `slurm/run_list.sh` inside an allocation (one `srun` per node).

**Data.** The three data sets of Table 1 are read from the following locations on NERSC:

- **AbacusSummit emulator suite**: `/pscratch/sd/n/ntbfin/emulator/hods/z0.5/yuan23_` (85 cosmology directories, `hod000`–`hod499`); cosmological parameters in `config/cosmologies.csv`. The catalogues measured are listed, one line per catalogue (cosmology, HOD, space, line of sight), in `config/runs_emulator*.txt` (8 HODs per cosmology), `runs_hod_calibration.txt` (72), `runs_local_*.txt` (240), `runs_extra.txt` (54) and `runs_real_space.txt` (27); the fiducial catalogue and the HOD-matched pairs are in `config/hod_selection.json`.
- **Official DESI mocks**: `/global/cfs/cdirs/desi/cosmosim/AbacusHOD_mocks/` (1786 files) and `.../CubicBox/LRG/z0.500/AbacusSummit_base_c000_p`
- **Quijote**: Rockstar halo catalogues `{set}/{i}/out_3_pid.list` for the sets `fiducial` (realisations 0–3999), `fiducial_ZA`, `Om_p/m`, `Ob2_p/m`, `h_p/m`, `ns_p/m`, `s8_p/m`, `Mnu_p/pp/ppp` (realisations 0–499), transferred with Globus from the Quijote public data to `/pscratch/sd/h/hugon/quijote/`, following the instructions of the Quijote documentation (`https://quijote-simulations.readthedocs.io`).

Intermediate products (measurements, trained emulators, chains; $\simeq$ 3 GB) are stored in `/pscratch/sd/h/hugon/astra-ds-store/` and are regenerated by the steps below.

**Steps.**

```
python scripts/select_hods.py; python scripts/
    select_runs.py; python scripts/
    select_local_runs.py
salloc ... bash slurm/run_list.sh config/runs_
    *.txt # measurements
sbatch slurm/night_stage1.sbatch # official
    mocks + Quijote
python scripts/build_emulator_dataset.py
python scripts/build_pair_dataset.py --holdout
    0 --out pair_dataset_full.npz
```

```
python scripts/train_emulator.py --nseeds 10 --
    pairs data/pair_dataset_full.npz --models-
    dir models_pairs --tag _pairs
python scripts/compare_methods.py --models-dir
    models_pairs --validation-tag _pairs --
    compress --cov covbox --fiducial official
    --params omega_b omega_cdm n_s sigma8 w0 wa
python scripts/quijote_fisher.py --params Om s8
     --rebin 2 --tag _p0c0; python scripts/
    summarize_quijote.py
python scripts/mcmc.py --data official:000:0 --
    vector ASTRA --cov covbox --volume-factor
    1; python scripts/summarize_mcmc.py
bash slurm/quijote_classes.sh # ASTRA classes
    on Quijote (run_quijote.py --classes)
python scripts/quijote_classes_fisher.py --
    params Om s8 --tag _p2; python scripts/
    classes_figures.py
python scripts/robustness.py; python scripts/
    plot_real_space.py
python scripts/paper_figures.py --params
    omega_b omega_cdm n_s sigma8 w0 wa
python scripts/paper_figures_v1style.py --
    params omega_b omega_cdm n_s sigma8 w0 wa
python scripts/readable_figures.py; python
    scripts/env_maps.py; python scripts/
    classes_figures.py
cd paper && make # article, article_fr,
    vulgarisation, poster
```

The full set of measurements costs $\simeq 200$ node-hours (1043 emulator-suite catalogues, 1786 small boxes and 25 base boxes, 11 000 Quijote realisations at $\simeq 5$ s each).

| Setting | Value |
|---|---|
| Subvolumes | $4^3$ of $500\,h^{-1}$Mpc |
| ASTRA padding, randoms | $30\,h^{-1}$Mpc periodic, $N_r = 4N_g$ |
| DS smoothing, mesh | Gaussian $10\,h^{-1}$Mpc, $5\,h^{-1}$Mpc CIC |
| Quartiles | 4, equal counts |
| Geometry randoms | $5N_g$ per subvolume |
| $s$, $\mu$ bins | $5\,h^{-1}$Mpc in $[0, 150]$, 120 in $[-1, 1]$ |
| Multipoles | $\ell = 0, 2, 4$ |
| Data vectors | $\ell = 0, 2$, $s \in [20, 150]$, $10\,h^{-1}$Mpc |
| Seed | 42 |

Table 12: Numerical settings.

# B Mathematical details

**Fisher matrix.** For a data vector $\boldsymbol{d}$ (the measured statistics), the Fisher matrix of the parameters $p = (\boldsymbol{\theta}, \boldsymbol{h})$ is

$$F = D^T C^{-1} D + P_h, \qquad D_{ai} = \frac{\partial d_a}{\partial p_i}. \tag{3}$$

Each ingredient has a role: $D$ (one column per parameter) says how much each element of the data vector moves when a parameter moves; $C^{-1}$ weights these movements by the inverse of the noise, so that a change counts in proportion to its significance, with the correlations between bins taken into account; $P_h$ adds the prior knowledge on the HOD parameters (here a Gaussian of the width of their uniform prior, on the HOD block only). $F$ is the inverse of the parameter covariance expected for an unbiased, optimal analysis: the marginalised errors are $\sigma_i = \sqrt{(F^{-1})_{ii}}$ (all other parameters free) and the errors at fixed other parameters $1/\sqrt{F_{ii}}$. Comparing two statistics amounts to comparing their $\sigma_i$.

**Hartlap factor.** The covariance is estimated from $N_s$ simulated samples. The inverse of an estimated covariance is biased: with noise, $\hat{C}$ has some accidentally small eigenvalues, so $\hat{C}^{-1}$ is on average too large, and the Fisher errors too small. For Gaussian data the bias is known exactly, and it is removed by the factor [14]

$$\hat{C}^{-1}_{\text{unbiased}} = \frac{N_s - N_d - 2}{N_s - 1}\,\hat{C}^{-1}, \tag{4}$$

where $N_d$ is the length of the data vector. The factor approaches 1 when $N_s \gg N_d$, and decreases quickly when $N_d$ approaches $N_s$: with 192 samples it is 0.72 for $N_d = 52$ but 0.31 for $N_d = 130$, which also increases the noise of the forecast. This is why we compress long data vectors, and why 1786 independent boxes (factor 0.99 after compression) are much better than 192 subvolumes.

**Compression.** $\boldsymbol{t} = W^T \boldsymbol{d}$, $W = \tilde{C}^{-1} D$, with $D$ the derivatives with respect to the 18 parameters and $\tilde{C}$ a Ledoit–Wolf shrinkage covariance: $\boldsymbol{t}$ has one element per parameter, each a weighted sum of the data designed to be as sensitive as possible to that parameter. If $\tilde{C} = C$, $\boldsymbol{t}$ keeps all the Fisher information of $\boldsymbol{d}$. If $\tilde{C}$ is only an approximation, the weights are not optimal and $\boldsymbol{t}$ keeps less information. This cannot bias the forecast towards optimism: $\boldsymbol{t}$ is a linear function of the data, its derivatives are $W^T D$ and its covariance $W^T C W$ is measured from the samples, so its Fisher matrix $(W^T D)^T (W^T C W)^{-1} (W^T D)$ is exact for $\boldsymbol{t}$, and it is always smaller than or equal to that of $\boldsymbol{d}$. A poor compression can only make the errors larger. The covariance of $\boldsymbol{t}$ is short (18 elements), so its Hartlap factor is close to unity.

**Noise of the derivatives.** Derivatives estimated as an average over $N$ pairs are $\hat{D} = D + \epsilon$, with a noise of covariance $\Sigma_D / N$. The standard Fisher estimator then has the expectation

$$\langle \hat{D}^T C^{-1} \hat{D} \rangle = D^T C^{-1} D + \mathrm{Tr}\big(C^{-1} \Sigma_D\big)/N. \tag{5}$$

The second term is a positive bias: the noise adds spurious information. It is roughly the same in all directions of parameter space, but matters only where the true information $D^T C^{-1} D$ is small, i.e. in the degenerate directions; the marginalised errors, set by the inverse of the matrix, are controlled precisely by these weak directions. We use three tests:

1. the noise fraction $\mathrm{Tr}(C^{-1}\Sigma_D)/(NF_{ii})$; above 20–30% the forecast is suspect;

2. the bracketing of [13]: the standard estimator is biased high; a compressed estimator, in which half of the simulations define the compression and the other half measure its information, is biased low. If both give the same error the noise does not matter; if they differ by factors 2–10 the result reflects the noise, not the physics. We call a forecast reliable when they agree within a factor 1.5, and quote their geometric mean, which is nearly unbiased;

3. convergence: the forecast is recomputed with $N/2$ pairs.

**Volume.** The forecasts are computed for one $(500\,h^{-1}\mathrm{Mpc})^3$ volume, where the absolute errors are comparable to the prior widths; this is the nature of a comparison of information content and is not corrected. The ratios $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ do not depend on the volume as long as the covariance scales as $1/V$ and the emulator error is negligible, which holds at this volume.

**Emulator loss.** With outputs standardised by the statistical error of a $2\,h^{-1}$Gpc box $\sigma_{\rm box}$,

$$\mathcal{L} = \Big\langle \frac{(f(x_i)-y_i)^2}{\sigma^2_{\rm box}} \Big\rangle + \lambda \Big\langle \frac{[(f(x_i)-f(x_j))-(y_i-y_j)]^2}{\sigma^2_{ij}+\epsilon^2} \Big\rangle \tag{6}$$

where the second average runs over the 2544 pairs of catalogues, $\sigma_{ij}$ is the error of their difference (64 paired subvolumes), $\lambda = 1$, $\epsilon = 0.05\,\sigma_{\rm box}$.

**Regression over pairs.** $\boldsymbol{d}_c(h) = \boldsymbol{a}_c + G\,V(h-h_0)$ for $c = c_\pm$, with a shared response $G$ and $V$ the 3 leading directions of the HOD space (singular vectors of the HOD response of the 72 fiducial catalogues); derivative $(\boldsymbol{a}_+ - \boldsymbol{a}_-)/(\theta_+ - \theta_-)$; errors by bootstrap over subvolumes.

**Inference.** Gaussian likelihood $-2\ln\mathcal{L} = (\boldsymbol{d} - \boldsymbol{f}(\boldsymbol{\theta},\boldsymbol{h}))^T (C/V_{\rm f} + C_{\rm emu})^{-1} (\boldsymbol{d} - \boldsymbol{f})$, with $C$ the official covariance, $V_{\rm f}$ the data volume in units of $(500\,h^{-1}\mathrm{Mpc})^3$ and $C_{\rm emu}$ the emulator error; uniform priors over the training range; `emcee` with 64 walkers and 8000 steps. Calibration is tested by the fraction of runs whose truth lies within $1\sigma$ of the posterior mean (68% expected).

# C Methodological path and intermediate results

This appendix details the steps of Table 5 in chronological order.

**1. HOD files are not realisations.** A first version of the pipeline averaged the statistics of 64 different HOD files per cosmology to build "covariance" and "derivative" sets, and measured them in the central $500\,h^{-1}$Mpc cube of each box. The HOD files are different HOD models, not realisations: their scatter ($\sim 55\%$) dominated the cosmological differences ($\sim 1\%$), and the finite differences were compatible with noise (median $|S/N| < 1$ per bin for $\omega_b$, $n_s$, $\sigma_8$). That version also had five implementation problems, all corrected: pair counts assumed periodic boundary conditions in a non-periodic cut box; the 2PCF used the mid-point line of sight from the centre of the box instead of the axis along which the RSD were applied, which erased the quadrupole; quartiles were computed jointly on galaxies and randoms, giving unequal and realisation-dependent sizes; the Delaunay triangulation was computed on the cut box without periodic padding; and ASTRA was first applied to the random catalogue alone, so that the random quartiles carried no information on the galaxy field and their correlations were consistent with zero.

**2. HOD-matched pairs.** We then selected, for each $\pm$ pair, the two catalogues with the closest HODs (twice closer than random pairs, Table 3) and differenced them subvolume by subvolume (Section 5.2). The differences are precise but dominated by the residual HOD mismatch: a 2% change of $\omega_b$ implied tens of per cent changes of the 2PCF at $s \sim 20\,h^{-1}$Mpc, and the Fisher matrices were singular.

**3. HOD calibration.** The HOD response measured on 72 fiducial catalogues and subtracted from the pairs was predicted with a 45% (linear) and 8% (GP) error; with an HOD term ten times the signal, even 8% leaves a residual of the size of the signal.

**4. Emulator.** The first emulator, trained on 8 HOD draws per cosmology, reproduced held-out cosmologies to a fraction of the statistical error, but did not capture the response to $\omega_{\rm cdm}$ below $70\,h^{-1}$Mpc (Figure 13). A preliminary comparison with this

emulator gave ASTRA and density split comparable overall, density split ahead in the voids (Q1) for $\omega_b$ and $\omega_{\rm cdm}$, and their combination 15–25% better than either.

**5. Densification.** 240 catalogues with HODs close to the fiducial were added in the cosmologies of the derivative grid; this alone did not stabilise the derivatives (70–170% scatter between networks).

**6. Regression over pairs.** Introduced as an emulator-independent estimator (Section 5.4); it revealed that the statistics respond essentially to one combination of the HOD parameters.

**7. Training on catalogue differences.** Halved the error of the predicted differences between held-out catalogues for $\omega_b$ and $\sigma_8$ (21%→10%, 13%→7%), down to their noise level (Table 4); the small-scale weight was dropped because it destabilised the auto-correlation forecasts. With the 192-subvolume covariance, emulator and regression then agreed on ASTRA being as good as or 0–20% better than density split (Tables 6 and 7).

**Learning curve.** To decide whether more training catalogues would reduce the emulator error, we retrained the emulator of the 2PCF and of the Q1, Q4 random-quartile cross-correlations on 25, 50, 75 and 100% of the training set. The median test error, in units of $\sigma_{\rm box}$, drops from 0.7–1.3 to 0.5–0.8 between 25 and 50%, but only to 0.5–0.7 at 100% (ASTRA Q1: 1.01, 0.59, 0.55, 0.51; density split Q4: 0.69, 0.51, 0.48, 0.46). The error has reached a plateau at $\simeq 0.5\sigma_{\rm box}$, comparable to the sampling noise of a single catalogue: doubling the training set ($\simeq 230$ node-hours) would improve it by $\sim 10\%$ at most, and we did not do it.

**8. Covariance.** With the 1786 independent boxes and the same derivatives, ASTRA's lead disappeared (Table 9).

**9. Quijote.** The forecasts were first computed for all six parameters. An automatic convergence test run overnight reported that the covariance was not converged, because it compared full forecasts in which the derivative noise dominates the change; once the two effects were separated, the covariance was found converged and the extra processing it had triggered was cancelled. The six-parameter forecasts were then found to depend on the derivative noise (standard and compressed estimators differing by factors 2–10), which led to the parameter subsets of Section 7.2. A first overnight result, ASTRA tighter than density split on $\sigma_8$ by a factor 2–3 and density split tighter on $\Omega_m$, $\Omega_b$, survives in the reliable subsets only for the auto-correlations and halo quartiles.

**10. MCMC.** The first inference runs on full boxes were limited by the emulator error; on one subvolume, they are limited by the prior (Section 7.4).

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